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HIGH AND RISING INSTITUTIONAL CONCENTRATION OF
AWARD-WINNING ECONOMISTS

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ABSTRACT

We analyze the academic affiliations of nearly 6,000 award-winning researchers in 18 major fields in the natural sciences, engineering, and social sciences from the 1820s to the 2020s, focusing on the 1960s onward. The analysis reveals a trend of declining concentration in the institutional affiliations of award-winning researchers, shifting from a few science-strong universities in high-income countries to a more diverse set of institutions across the world. The decline in concentration is observed in all fields except one: economics. The institutional affiliations of prizewinning economists have become more concentrated over time, making economics the most concentrated field. We associate the higher concentration of prizewinning work in economics with the field's stronger sorting by institutional prestige, its lower reliance on specialized equipment and instruments, and its assessment of findings based on a synthesis of evidence rather than on decisive experiments or proofs. We discuss the benefits and costs of this high and rising institutional concentration of prizewinning economists.

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Scientific research has spread from being concentrated in a few universities and research centers in advanced Western countries to other institutions in those countries and in other countries around the world. The wider geographic distribution of researchers has been spurred by the expansion of higher education worldwide (Schofer and Meyer, 2005); the rise of team research (Jones, 2021; Wu et al., 2019; Wuchty et al., 2007; Adams et al., 2005); the spread of global research collaborations as emerging economies have improved their scientific capacities (Coccia and Wang, 2016; Xie et al., 2014; Adams, 2013; Wagner and Leydesdorff, 2005); and the Internet making it easier for teams to function across different locations (Teasley and Wolinsky, 2001).¹

The spread of research to more institutions occurred within countries and globally. In the United States, the number of universities awarding PhDs in scientific disciplines increased from 171 in 1960 to 331 in 1980, and to 457 in 2022 (NSF, 2006, 2023). Globally, the number of higher education institutions jumped from 3,701 in 1960 to over 8,500 in 1980, and then to more than 21,000 in 2022, with increases seen in virtually every region of the world (IAU, 1959, 2022), adding huge numbers of potential institutional addresses on research papers.² The broader geographic distribution of knowledge production enhances the accessibility and dissemination of scientific knowledge to persons in both the established and newer research-producing countries, with potential spillovers to technological progress and economic growth.

While the number of academic centers with researchers has expanded greatly, it is less clear that the location of leading researchers, whose novel findings or answers to outstanding questions move the frontier of knowledge, has expanded commensurately. One possibility is that big breakthroughs come from established centers with a high concentration of leading researchers who interact positively with each other or with each other’s students, while researchers at newer institutions play a more limited role in advancing the scientific frontier.³ The opposite possibility is that the expanding number of locations doing research changes

¹Contrary to fears of brain drain, the increased number of international students has facilitated cross-country work, as both those who return to their country of birth and those who immigrate to another country are especially likely to engage in international collaborations (Horta et al., 2021; Freeman, 2015; Scellato et al., 2015; Kerr, 2008).

²The Web of Science shows a large increase in unique institutional addresses from 1980 to 2022 in Asian countries, with increases from 108 to 1,674 in China, from 31 to 308 in South Korea, from 437 to 708 in Japan, and from 392 to 1,360 in India.

³Past research has found that scientific production is positively correlated with faculty quality (Waldinger, 2010), peer quality (Waldinger, 2012), and superstar spillover (Azoulay et al., 2010). However, Kim et al. (2009) found that the research-productivity premium from an elite affiliation in economics and finance had largely disappeared by the 1990s, which makes the persistence of prize concentration harder to attribute to spillovers alone. Being in an elite group also increases the likelihood of obtaining research grants and fellowships and of publishing in top journals (Card and DellaVigna, 2020; Bol et al., 2018; Colussi, 2018; Kocher and Sutter, 2001; Hodgson and Rothman, 1999). This suggests that such groups have value, at least to their members.

the locus of scientific leadership, with researchers affiliated with new or historically less prestigious universities or centers increasing their share of scientific breakthroughs and their impact on the trajectory of scientific work.⁴

These considerations motivate our investigation of whether the spread of research to more universities and research centers has produced a comparable widening of the institutional affiliations of *leading researchers*. Given that science has historically used scientific prizes or awards rather than money (Merton, 1968) to distinguish the “best and brightest” (though many awards have monetary emoluments), we identify leading researchers as those who have won major scientific awards. We obtain data on the educational and professional affiliations of nearly 6,000 recipients of 171 scientific prizes in 18 scientific fields, including natural sciences, engineering, and social sciences.⁵ Our prize list ranges from famous awards such as the Nobel Prizes, the Fields Medal in mathematics, and the new \$3 million Breakthrough Prizes to the field-specific prizes that more narrowly defined fields give to their top researchers. The analysis covers the earliest scientific awards to the present, with particular attention to the post-1960 period when research activity expanded the affiliations on scientific papers to new institutions in all countries, including many that conducted little science in earlier years.

The paper is divided into three sections. Section 1 describes the data on prize recipients and their institutional affiliations and documents our key finding: the concentration of affiliations of prizewinning researchers fell in all fields except for economics, whose rising concentration made it number one in the institutional concentration of prizewinners. Section 2 examines three factors that appear to contribute to economics deviating from the other sciences: the greater impact of institutional prestige in hiring networks in economics; the limited role of physical capital in economic research, which allows leading universities to spend their money on researchers rather than on large labs and pieces of equipment; and the nature of evidence that makes it difficult for someone outside the mainstream to gain attention for novel findings that challenge consensual thinking. Section 3 evaluates the benefits and costs of the institutional concentration of prizewinners for progress in a field.

⁴The continued dominance of elite echelons in science could discourage new researchers from analyzing non-mainstream topics within their disciplines or in other sciences (Hudson, 2013) and may hinder the creativity and originality of scientists outside the elite group (Heckman and Moktan, 2020).

⁵Natural sciences include mathematics, physics, chemistry, life sciences, astronomy, and earth sciences; engineering fields include electrical, civil, energy, environmental, materials, and mechanical engineering; and social sciences include economics, political science, sociology, law, education, and psychology.

1 Institutional affiliation of prizewinning researchers

We begin by identifying a large set of scientific awards that academic societies and groups bestow on researchers to acknowledge ground-breaking innovations and contributions to science (Borjas and Doran, 2015), the receipt of which invariably raises the winners’ scientific prestige and ability to influence the frontier of knowledge and public policies (Ma and Uzzi, 2018). We examine a large set of awards, determined by different groups under different decision procedures, to allow us to generalize about the affiliations of prizewinners without delving into the decision-making processes by which groups give their awards.

Our initial expectation was that the expanding number of academic institutions in research would reduce the concentration of affiliations of awardees in all fields, albeit at differing rates depending on the varying importance of theory, physical capital, and research funding among fields. The data show that this is the trend in all fields except for economics.

To obtain our sample of awards, we started from three published lists of prestigious international academic awards (Zheng and Liu, 2015; Jiang and Liu, 2018; Meho, 2020).⁶ We modified these lists to include prizes established relatively recently that have gained high prestige, such as the Breakthrough Prizes; and exclude prizes that were awarded with a strong preference for nationality and/or region, such as the TWAS prize for developing countries, and awards for commercializing technology, such as the World Technology Award. The final dataset comprised 171 awards, with the earliest dating back to the 1840s. Our primary focus is on awards given from 1960 through 2022, when the number of researchers and research institutions ballooned worldwide.

We end up with 5,859 prizewinning scientists.⁷ Appendix Table OA1 gives the full list of awards and their scientific fields, links to their official websites, and the number of recipients of each award.

We obtain the academic affiliations of prizewinners at the time they won the award, when they earned their PhD degree, and annually throughout their careers.⁸ We follow affiliations throughout their whole careers because the affiliation at the time of the award may give a misleading picture of where the prizewinning work was done. For this measure, we

⁶These studies constructed their lists using tiered-checklist methods and surveys distributed to experts in each profession. Jin et al. (2021) assemble a related cross-disciplinary dataset of scientific prizes to study how prizes shape the growth of research topics.

⁷We count individuals awarded multiple prizes in the same discipline once but count recipients of multiple prizes in different disciplines as separate entries in their respective fields. There are 1,182 prizewinners who won multiple awards in a single discipline and 203 who won multiple awards across disciplines.

⁸We obtain data on their affiliations year by year from addresses on each scientist’s yearly publication, supplemented with manual searches for affiliations in years without any documented publications. We also relied on the descriptions of the award-winners on the official website of the award, Wikipedia, and the prizewinners’ scholarly homepages and university faculty pages.

construct a year-by-year panel and count *affiliation-years*—one observation for each year a prizewinner was affiliated with a given institution—so that institutions are weighted by how long they held prizewinners rather than by how many passed through.⁹ We treat the PhD institution as an indicator of the job network with which a researcher began their career and of scientific connections throughout their life, rather than as part of their career affiliation or as where they made their prizewinning contribution, though for some, the doctoral work laid its foundation. All three measures give, as we shall see, broadly comparable results about the divergence of economics from other fields.

Finally, we turn the affiliations of researchers in a field into a field-level measure of the concentration of affiliations of all prizewinners. We measure concentration by the Herfindahl-Hirschman Index (HHI), which is widely used in industrial organization and elsewhere to indicate whether a particular variable (sales or production in an industry, income or wealth of individuals, or prize awards in our case) is dominated by a small number of units or is more broadly distributed. The key input in our analysis is $p_{i,t}$, defined as the proportion of award-winning researchers affiliated with institution i among N institutions with at least one prizewinner in year t . The HHI for a given field with N institutions is then

$$HHI_t = \sum_{i=1}^N p_{i,t}^2 \in \left[\frac{1}{N}, 1 \right]. \quad (1)$$

The highest HHI is achieved when all prizewinners have the same affiliation, and it has a minimum value of $1/N$ if each of the prizewinners has a different affiliation. Given the dependence of the HHI on the number of institutions, we follow the common practice of scaling the HHI to have a minimum value of 0 using the normalization formula:

$$HHI_t^* = \frac{\sum_{i=1}^N p_{i,t}^2 - 1/N}{1 - 1/N} \in [0, 1]. \quad (2)$$

Following the practice for reporting HHI in percentage-point-squared terms, we move the decimal point of the calculated statistic by 10,000.¹⁰

Figure 1 shows the trends in HHIs for the affiliations of prizewinners over their careers across 18 fields in the natural sciences, engineering, and social sciences, covering the years

⁹Appendix Table OA2 lists the top 10 institutions in each field by the number of recipients; Appendix Table OA3 gives the complete ranking for economics by affiliation-years. A researcher holding a joint appointment contributes an affiliation-year to each institution. Such years are uncommon, ranging from 0.2 to 9.4 percent of observations across fields and only 1.0 percent in economics, and splitting them evenly between institutions changes no statistic we report by more than one percentage point.

¹⁰This conversion is widely used in antitrust analysis; see, e.g., the disclosure practices of the US Department of Justice [here](#).

up to 2022.¹¹ The sample for the natural sciences before 1825 is excluded due to the small number of awards during that period. To highlight the changes over time, we measure the HHI as ten-year centered moving averages. The trend in HHI is uniformly downward in the natural sciences from 1850 to 2022, though at differing rates and with increases in concentration during some periods. The trend is also uniformly downward in the engineering fields, where prizes began at a later date, and in the social sciences, save for economics, whose concentration trends upward.¹²

Table 1 displays the values of the HHI for the 18 fields over three periods: the year in which we calculated the first HHI (given in parentheses); 1960, which marks the start of the period during which most of our prizes were awarded; and 2022. Economics stands out as the sole field for which the concentration of affiliations increased over time and as the field with the highest HHI in 2022—a position it did not hold initially or in 1960 when it had an “average concentration,” ranked 9 out of 18, from highest (1) to lowest (18).

To determine whether the high concentration of university affiliations is limited to the institutions where the prizewinner spent their career or extends to the universities where they were affiliated when receiving the award and the institutions that granted their PhD, we compare the HHI rankings of the fields using all three measures. Figure 2 shows that the measures are positively associated, with economics being the most concentrated in all the metrics. Concentration at the time of the award is more closely correlated with concentration over the prizewinners’ careers than with concentration among PhD-granting institutions. Researchers move early, away from the department that trained them, often to a postdoctoral position or first job elsewhere, and then settle: in every field, more than 70 percent of prizewinners received their award at the institution where they spent most of their career, and in economics, 80.5 percent did.

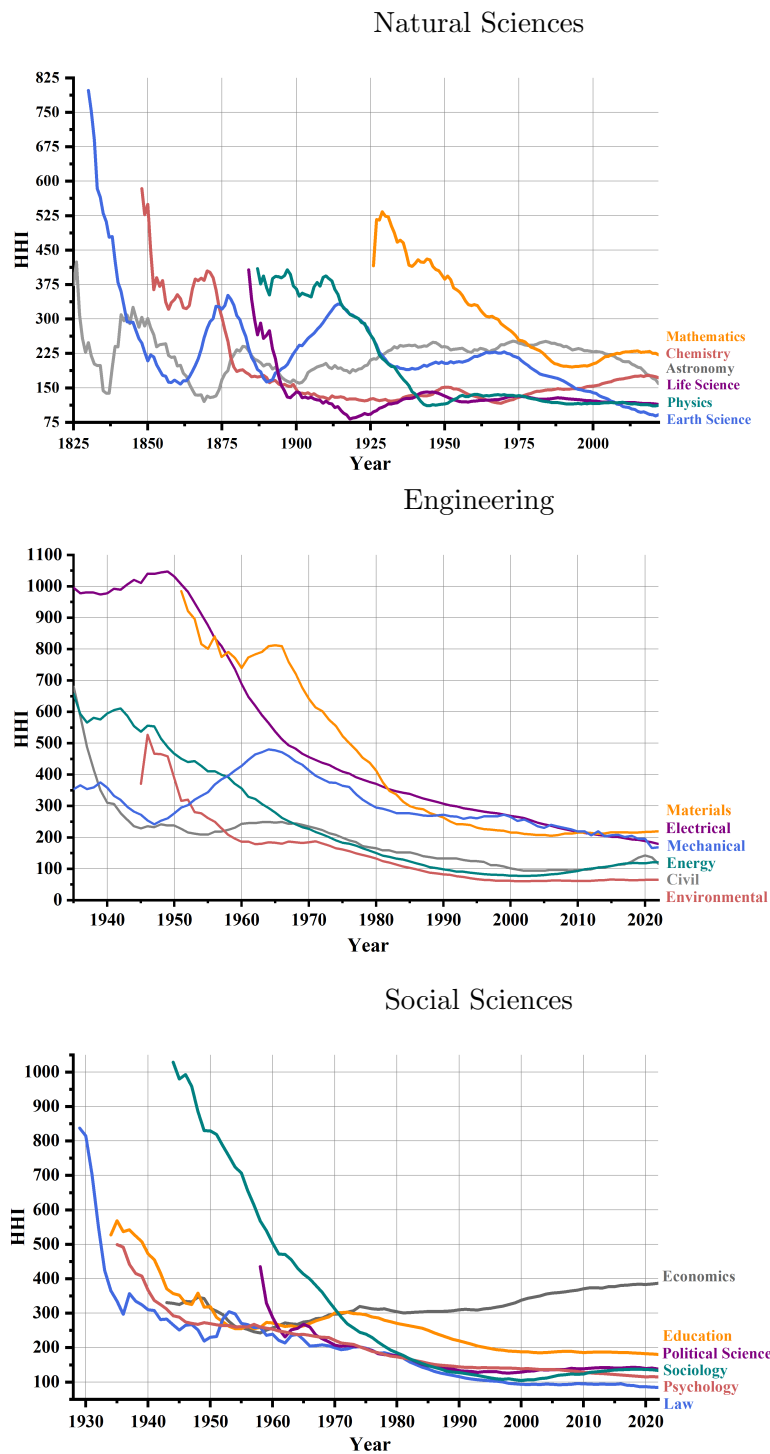
Because US universities employ a disproportionate number of top researchers, the institutional concentration of prizewinners could be driven by differences between universities in the US and universities in other countries.¹³ Appendix Figure OA5 shows that the finding holds when we restrict the sample to US-based institutions, even though US-based researchers

¹¹Appendix Figure OA1 plots the series for each field separately. Figure OA2 repeats the calculation using doctoral as well as employment affiliations, Figure OA3 restricts the sample to years after 1950, and Figure OA4 dates each affiliation at the year of the award.

¹²Since economics is part of the broader area of business studies, which gives prizes to other subfields, we compared the level and trend in the concentration of prizes in economics and the associated field of finance versus other business school disciplines, including accounting, management, marketing, and strategy. Economics and finance show high and increasing levels of concentration, whereas other business school subjects show low and decreasing levels of concentration (see Appendix C).

¹³Das et al. (2013) raises this concern about the US impact on the global distribution of citations; Krauss (2024) and Von Zedtwitz et al. (2025) document the large share of Nobel Prizes going to US-addressed researchers; and Nature’s 2024 editorial calls for a wider geographic distribution of the prizes.

Figure 1: Normalized employment HHI by prizewinner's affiliation over their career



Notes: Ten-year centered moving averages. Natural sciences are shown only from 1825 because of small sample sizes before then.

Table 1: Institutional concentration of prizewinners, based on total years of affiliation over their career: earliest year, 1960, and 2022

Academic field	Earliest HHI	1960 HHI	2022 HHI
Natural Sciences			
Mathematics	415.67 (1926)	331.16	222.09
Physics	409.83 (1887)	135.51	111.70
Chemistry	583.88 (1848)	131.97	172.02
Life Science	407.43 (1884)	120.94	114.03
Astronomy	3648.39 (1811)	232.76	159.19
Earth Science	797.97 (1830)	218.38	91.36
Engineering			
Electrical Engineering	967.82 (1891)	688.50	179.16
Civil Engineering	1289.25 (1930)	243.19	115.71
Energy Engineering	1479.29 (1921)	356.43	120.96
Environmental Engineering	370.23 (1945)	185.91	64.96
Materials Engineering	984.70 (1951)	739.88	219.64
Mechanical Engineering	554.10 (1896)	428.12	168.57
Social Sciences			
Economics	330.34 (1943)	257.34	386.99
Political Science	435.59 (1958)	287.51	137.66
Sociology	1028.84 (1944)	503.08	133.83
Law	837.50 (1929)	239.43	84.26
Education	526.63 (1934)	271.69	179.99
Psychology	499.44 (1935)	253.50	115.22

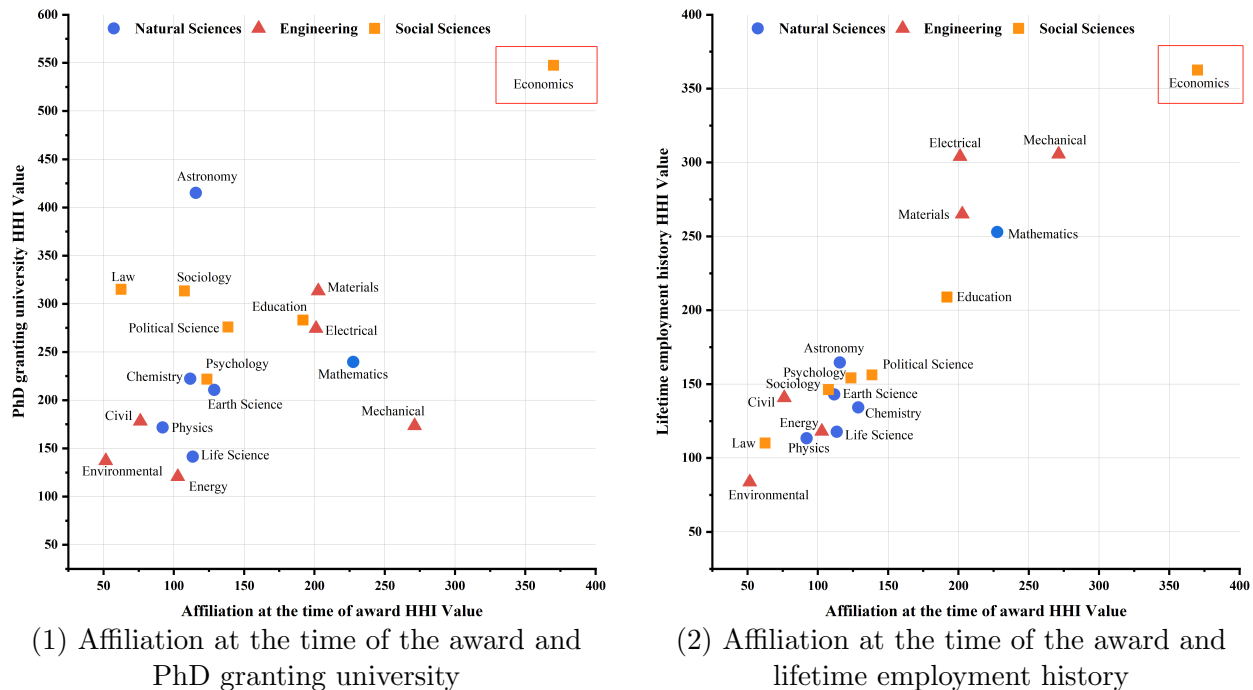
Notes: This table presents the normalized HHI of prizewinners based on affiliation over their careers weighted in (1) the earliest recorded year (years in parentheses), (2) 1960, and (3) 2022.

made up 77 percent of prize recipients in economics compared with 48 percent in physics, 54 percent in chemistry, and 61 percent in life sciences.¹⁴

Other measures of the concentration of affiliations, such as the Gini coefficient that economists use to assess income inequality, give a similar picture of economics as the “odd man out,” with prizewinners concentrated in a few institutions. The Gini coefficient for institutional affiliation over a lifetime is a massive 0.812 in economics (Appendix Table OA4)—a statistic that reflects the remarkable fact that 50% of all economics affiliation-years were spent at just 8 of the 249 institutions at which any prizewinner spent time: Harvard (10.9%), Chicago (8.0%), MIT (7.6%), Stanford (6.4%), Princeton (4.9%), Yale (4.3%), Berkeley (4.0%), and Columbia (3.3%). The top 27% of institutions account for 90%

¹⁴These numbers are close to the results of Krauss (2024) for Nobel Prizes, where US-based researchers received 41 percent of physics awards, 45 percent of chemistry awards, 47 percent of medicine awards, and 78 percent of economics awards from 1901 to 2022.

Figure 2: Institutional concentration of prizewinning researchers at the time of award, PhD granting university and lifetime employment history



Notes: Affiliation at the time of the award is the institution with which the individual was associated during the year the award was conferred. The PhD granting institution is the university where the individual obtained their doctoral degree. Lifetime employment history counts affiliation-years, one for each year the individual was employed at a given institution; years of education and doctoral training are excluded.

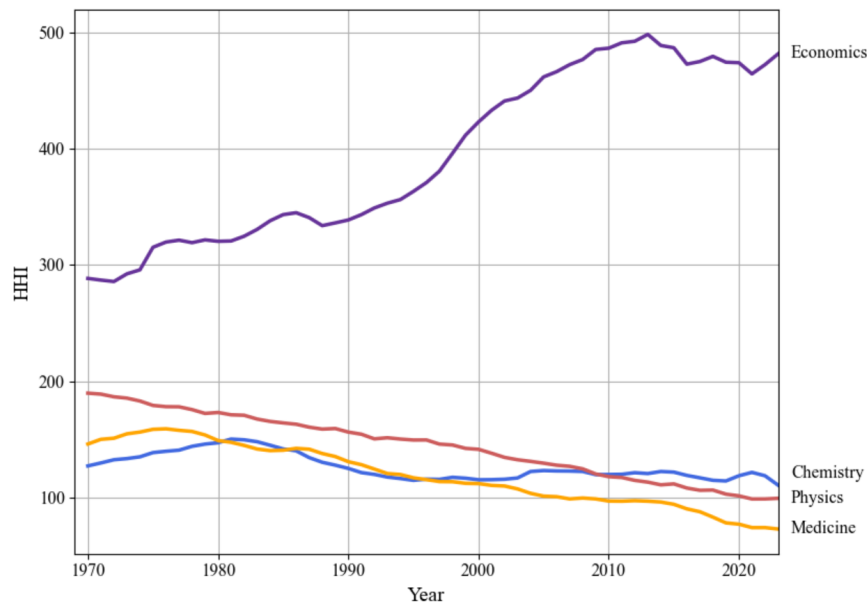
of prizewinners' affiliation-years in economics, compared with the top 37% in physics and the top 35% in chemistry and life science.

Finally, we examine the robustness of the finding that economics has a high and uniquely increasing concentration of affiliation. We use two other plausible measures of leading researchers, i.e., economists who win the Nobel Prize and economists who obtain a sufficient number of citations to be on Clarivate's list of Highly Cited Researchers (Clarivate, 2023). The Nobel Prize can be viewed as the ultimate top-down award, made by committees of experts who consult with other experts. Clarivate's use of citations can be viewed as a bottom-up "award" based on researchers who "vote" with their citations for research papers by authors with different affiliations.¹⁵

Both measures confirm the high level and rising trend in the concentration of the affiliation of leading economists compared to the declining trend in the affiliation of leading

¹⁵Clarivate determines its list based on the premise that each researcher "has authored multiple Highly Cited Papers that rank in the top 1% by citations for their field(s) and publication year in the Web of Science over the past decade ... refined using qualitative analysis and expert judgment."

Figure 3: Institutional concentration of Nobel laureates from 1969 to 2022



Notes: Ten-year centered moving averages. A field’s institutional concentration in year t is calculated by the HHI of year t academic affiliations of Nobel laureates up to year t as well as those after year t . 1969 was selected as the starting year, marking the inaugural award in economics.

researchers in other disciplines. Figure 3 shows the concentration of affiliations over the careers of Nobel laureates in economics, chemistry, medicine, and physics from 1969 to 2022. Appendix Figure OA6 extends the analysis to the concentration of PhD granting institutions and the institutions with which they were affiliated when they obtained their awards, yielding the same results. Appendix Figure OA7 shows the HHI of Clarivate’s Highly Cited Researchers by affiliation in the Web of Science from 2014 to 2025. Economics exhibits the highest concentration under both measures.

In summary, economics is the anomaly, showing an increasing trend in the concentration of affiliations of leading researchers despite the expansion of scientific research to more institutions and more countries than ever before. What explains the anomaly?

2 Mechanisms

This section examines three factors associated with the concentration of awards in economics. We use the word associated because these factors can be viewed as a manifestation of the anomaly or as an exogenous independent cause in a causal model that goes beyond the scope of our analysis. For each factor, we begin with the prizewinners and then ask whether the pattern holds in the academic labor market as a whole, since concentration is itself an

outcome of how that market allocates talent.

The first factor is the role of institutional prestige in academic affiliation. Award-winning economists are affiliated with a small number of highly ranked universities throughout their careers, from graduate school to their first jobs—more so than prizewinners in any other field, including the other social sciences. The same sorting appears in the broader academic market: jobs at prestigious economics departments are concentrated among graduates of a few top-ranked programs, more so than jobs at prestigious departments in other fields. These patterns are consistent with the view that prestige plays a relatively stronger role in economics than in other fields and may contribute to the field’s high level of institutional concentration.

The second and most likely candidate for a causal factor is physical capital. Economics research requires relatively little expensive physical capital compared to most of the natural sciences and engineering, and this difference can account for the divergence between economics and those fields in the institutional concentration of award-winners.¹⁶ On the supply side, economists can move between institutions at lower cost than researchers who depend on specialized laboratories, instruments, or large shared facilities. On the demand side, wealthy, prestigious institutions that seek to hire leading economists can do so without making large investments in buildings or equipment. The result is that mobility among institutions is much higher in economics than in most other research fields, both among prizewinners and in the broader academic labor market.

The third factor is the nature of the evidence that underpins potential prizewinning discoveries in economics. In economics, many significant contributions arise from a synthesis of evidence around key claims of the mainstream theory rather than from definitive experimental designs or mathematical proofs that either succeed or fail unequivocally, supporting or overturning what most experts previously believed. It is easier for researchers outside the mainstream to establish recognition through clear and demonstrable results—such as the efficacy of a medicine or to prove or disprove a mathematical conjecture—than in a field that lacks such definitive tests. This difference makes institutional affiliation more consequential in economics than in most natural sciences, engineering, and mathematics.

The evidence on prestige spans all 18 fields, and in that dimension, economics stands apart from the other social sciences as clearly as it stands apart from the natural sciences and engineering. Physical capital and the nature of evidence set economics apart from the natural sciences, engineering, and mathematics, but not from the other social sciences, which require no more specialized equipment than economics and whose findings do not live or die

¹⁶It does not, however, set economics apart from the other social sciences, or from mathematics prior to the success of AI in solving major math problems.

on decisive experimental tests. But, as we have seen, while the institutional concentration of prizes in economics has risen, that in the other social sciences has fallen at roughly the same rate as in all other fields – an issue which we will come back to at the end of the section.

2.1 Prestige and assortativity

Institutions with stronger reputations attract more funding, more talented students, and other resources that sustain their standing (Nowogrodzki, 2022; Wapman et al., 2022; Clauset et al., 2015; Petersen et al., 2014; Merton, 1968). To see whether institutional rankings and prestige attract stars more in economics than in other fields, we compare the average rankings of award-winners’ doctoral and employment affiliations.

We restrict the analysis to US-based affiliations after 1960 and use the 2024 *US News Best Colleges* rankings of national universities as our benchmark. We prefer a national ranking because global rankings weight research output, reputation surveys, and internationalization differently from one provider to the next.¹⁷ Although general university rankings do not fully capture field-specific strength, they provide a common benchmark for comparing the broader institutional prestige associated with prizewinners across disciplines.¹⁸

Figure 4 compares the affiliation rankings of 18 academic fields. The results show that award-winning economists are, on average, affiliated with higher-ranked universities than prizewinners in any other field, both at the PhD stage and in subsequent employment. Award-winning economists are affiliated with the highest-ranked universities of any field at both stages: their doctoral institutions average a rank of 13, and their subsequent employers a rank of 15. In other fields, by contrast, doctoral institutions typically average between 20 and 35, and employment affiliations are mostly between 20 and 40.¹⁹ These patterns are consistent with the view that award-winning economists are more concentrated in higher-ranked universities and that institutional prestige may play a stronger role in economics than in other fields.

We next extend the analysis beyond the small subsample of prizewinners to the broader academic hiring market, where researchers are sorted across institutions. Differences in how these markets allocate academic talent may contribute to differences in institutional concentration across fields. We use the dataset from Wapman et al. (2022), which provides

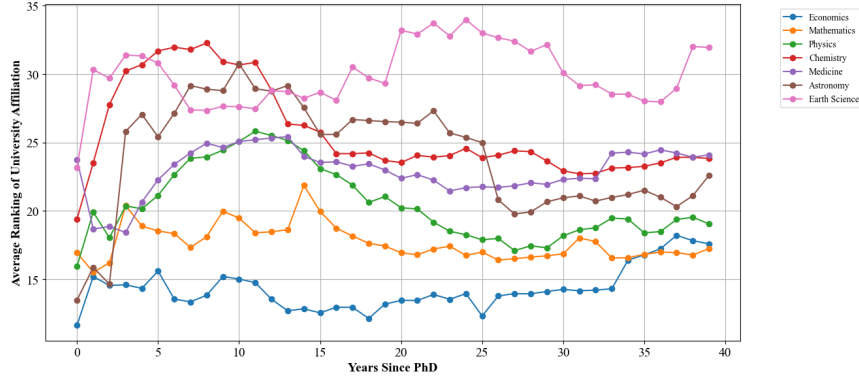
¹⁷ *US News* has ranked institutions beyond the top 25 only since 1996, so we apply a single recent cross-section rather than year-specific rankings. Positions at the top are stable over time, which aligns with our focus on prizewinners at highly ranked institutions.

¹⁸ Appendix Table OA5 repeats the comparison using two global rankings, and our results are stable.

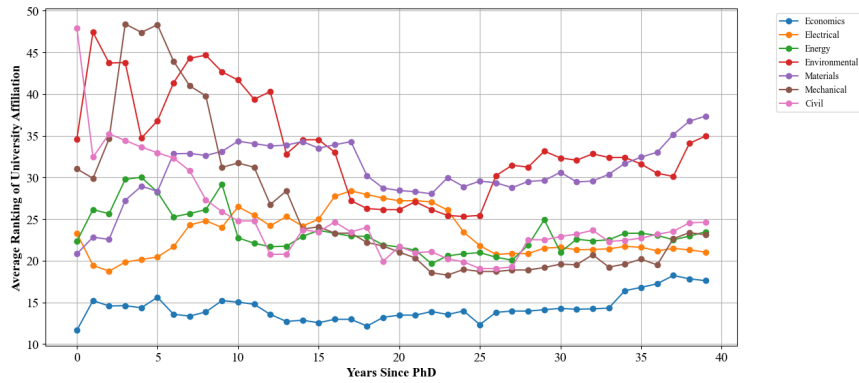
¹⁹ Comparing average rankings at the doctoral and employment stages does not show whether individual prizewinners are at similarly ranked institutions at both stages, since large individual differences can offset in the aggregate. Appendix Table OA6 compares each prizewinner’s doctoral and employment institution, and economics has the highest share of those whose two institutions are similarly ranked.

Figure 4: Average university rankings

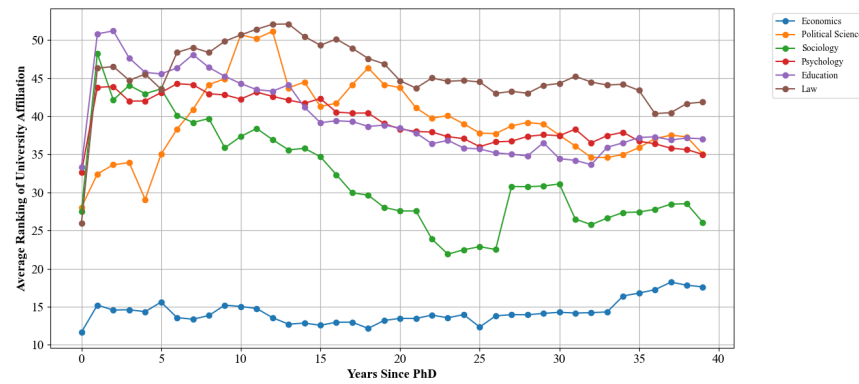
Natural Sciences



Engineering



Social Sciences



Notes: Average institutional rankings of prizewinners' affiliations over time. The sample includes US-based affiliations post-1960. Rankings are benchmarked to the 2024 US News Best National University Rankings. Year 0 refers to the PhD-granting institution, and subsequent years indicate the rankings of institutions where individuals are employed. Lower numerical values indicate higher institutional prestige.

detailed information on the academic employment and doctoral education of 295,089 tenure-track faculty across 10,612 departments at all PhD-granting US universities between 2011 and 2020. We aggregate the 107 subfields in the original data into 18 broader fields to align the hiring-market analysis with the rest of the paper; the full mapping is reported in Appendix Table OA7. The dataset also provides subfield-specific institutional rankings based on SpringRank, using observed hiring and placement networks.

Wapman et al. (2022) show that economics and finance are among the fields with the lowest rates of upward mobility in faculty hiring. As a complementary network-based characterization of mobility, we calculate the correlation between the ranks of the PhD-granting and employing institutions. The higher the correlation, the greater the tendency for people from top PhD institutions to get jobs in top-ranked employing institutions. The correlation for economics places it among the most assortative fields, indicating strong sorting between institutions of similar rank.²⁰

We further examine the concentration of hiring networks to see whether that concentration is primarily at the very top of the market or extends to all institutions within each field. We define hiring concentration as the normalized HHI of the PhD-granting institutions from which a university’s current faculty received their doctorates, multiplied by 10,000 for consistency. Using SpringRank within each field, we calculate this measure for the top 10, 30, 50, 100, and all institutions. To quantify the rate at which concentration decreases across institutional ranks, we estimate a power-law distribution that relates the concentration of awards to the rank of universities, and we use the estimated negative coefficient as a summary measure. The more negative the coefficient, the more concentrated awards are at highly ranked institutions.²¹

Table 2 gives the estimated power-law coefficients and the HHI. In the power-law specification, the estimated coefficient for economics is -0.627 , the largest in absolute value among the 18 fields. Economics exhibits the highest hiring concentration at the top 10 and top 30 levels, but its ranking in concentration falls along the distribution: ranking second at the top 50 level, fourth at the top 100 level, and seventh when the full academic market is considered.²² High hiring concentration in economics is concentrated among top-ranked institutions rather than throughout the broader academic market.

One interpretation of this pattern is that economics sorts at the entry margin. If its strongest departments kept more of their PhDs in academia than the strongest departments

²⁰See Appendix Table OA8 for the mobility and assortativity results across all 107 fields. This matches Terviö (2011), who finds that economics departments cluster into groups that rarely hire across cluster lines; among top mathematics departments, by contrast, hiring patterns are consistent with random matching.

²¹The power-law approximation is given by $f(x) = \beta x^{-\alpha}$, where $-\alpha$ denotes the power-law coefficient.

²²See Appendix Table OA9 for hiring concentration results across all 107 subfields.

Table 2: The academic hiring market by concentration

Academic Field	# of schools	Power-law coefficient	HHI concentration perspective				
			Top 10	Top 30	Top 50	Top 100	All
Natural Sciences							
Mathematics	224	-0.418	399.08	300.12	238.99	169.52	102.60
Physics	221	-0.497	462.38	313.35	265.72	181.17	120.02
Chemistry	229	-0.520	638.58	333.91	267.04	190.72	110.05
Life Sciences	282	-0.226	186.70	132.40	105.21	79.17	61.62
Astronomy	94	-0.465	328.86	227.65	175.64	–	127.25
Earth Sciences	191	-0.419	342.43	194.90	161.16	127.13	98.70
Engineering							
Electrical Engineering	239	-0.605	766.86	387.70	269.99	177.58	116.48
Civil Engineering	154	-0.542	432.13	293.61	233.63	177.99	151.32
Energy Engineering	142	-0.384	349.54	177.31	142.25	117.47	104.22
Environmental Engineering	183	-0.377	342.89	178.77	146.93	115.73	100.82
Materials Engineering	90	-0.542	418.87	245.26	186.03	–	130.24
Mechanical Engineering	180	-0.480	606.15	330.74	266.34	195.83	157.45
Social Sciences							
Economics	204	-0.627	777.68	455.91	272.02	189.34	121.29
Political Science	205	-0.514	719.23	424.25	349.98	241.88	161.25
Sociology	194	-0.319	392.86	330.78	266.37	186.85	135.52
Law	79	0.254	126.92	105.32	101.88	–	109.90
Education	202	-0.130	381.72	213.08	171.48	127.21	90.28
Psychology	242	-0.242	211.68	183.18	137.07	108.23	69.85

Notes: “Number of schools” is the count of distinct hiring institutions in the dataset. We estimated the power-law coefficient by regressing the log of the normalized HHI on the log ranking of institutions. The columns ‘Top 10’ to ‘All’ represent the HHI for the corresponding top-ranked institutions.

in other fields, they would supply an outsized share of academic hires, and the concentration at the top would reflect entry into academia rather than allocation within it. Using education and employment histories from Revelio Labs to examine the academic placement rate of PhD programs,²³ we find that the opposite is true: the probability of an academic placement falls 1.9 percentage points for every ten percentiles up the program ranking, estimated against the other seventeen fields as the baseline.²⁴ The leading economics departments place a smaller

²³Revelio Labs assembles career histories mainly from LinkedIn profiles. Our sample includes US PhD recipients in 18 fields from the 1995–2015 cohorts: coverage is thin before 1995, and ending in 2015 lets us observe every cohort ten years out, by which point careers have settled. Program rank is again the within-field SpringRank of Wapman et al. (2022). LinkedIn under-represents tenured faculty, so measured academic shares fall below administrative counts; however, there is no reason to expect the under-representation to be field-specific, so it should shift levels rather than the ordering. The estimating sample includes 267,421 individuals with an observed first job, of whom 11,408 are economists, and 258,352 observed ten years out, of whom 10,980 are economists.

²⁴Appendix Table OA10 reports the estimates. Panel A gives the within-economics slope on the raw rank scale: a graduate of the highest-ranked program is 19.1 percentage points less likely to be in academia ten years after the PhD than a graduate of the lowest-ranked one. Panel B places all eighteen fields on a common within-field percentile scale; the pooled slope for the other seventeen is essentially zero at ten years, against -1.94 for economics.

share of their graduates in academia than lower-ranked ones do, and the top of the academic hiring ladder is concentrated in those departments even so.

2.2 Physical capital

Scientific discoveries in the natural sciences and engineering often depend on experiments. This creates incentives for researchers to locate at institutions with the resources needed to produce such evidence. Heavily experimental fields depend on specialized laboratories, as in chemistry and the life sciences; in physics and astronomy the need extends beyond the department to large, publicly funded facilities such as accelerators and observatories (Nagaraj et al., 2020; Iaria et al., 2018; Helmers and Overman, 2017). The reliance on physical capital may affect researcher dispersion across institutions, as no single institution can support the full range of infrastructure required for all types of work. In contrast, in less equipment-intensive fields such as economics (and other social sciences), a single institution can support a broader range of research activity, allowing top researchers to be concentrated within a smaller set of institutions. We test the role of physical capital in explaining the difference in institutional concentration between economics and the natural sciences and engineering by comparing researcher mobility across fields. Scholars in experimental fields face higher costs of moving across institutions, since their work depends on specialized equipment and facilities, so mobility should be lower in those fields than in economics.

We estimate a linear probability model of annual mobility among prizewinners:

$$y_{it} = \sum_{f=1}^{18} \beta_f \mathbf{1}\{Field_i = f\} + \phi_{e(i,t)} + \psi_t + \varepsilon_{it}, \quad (3)$$

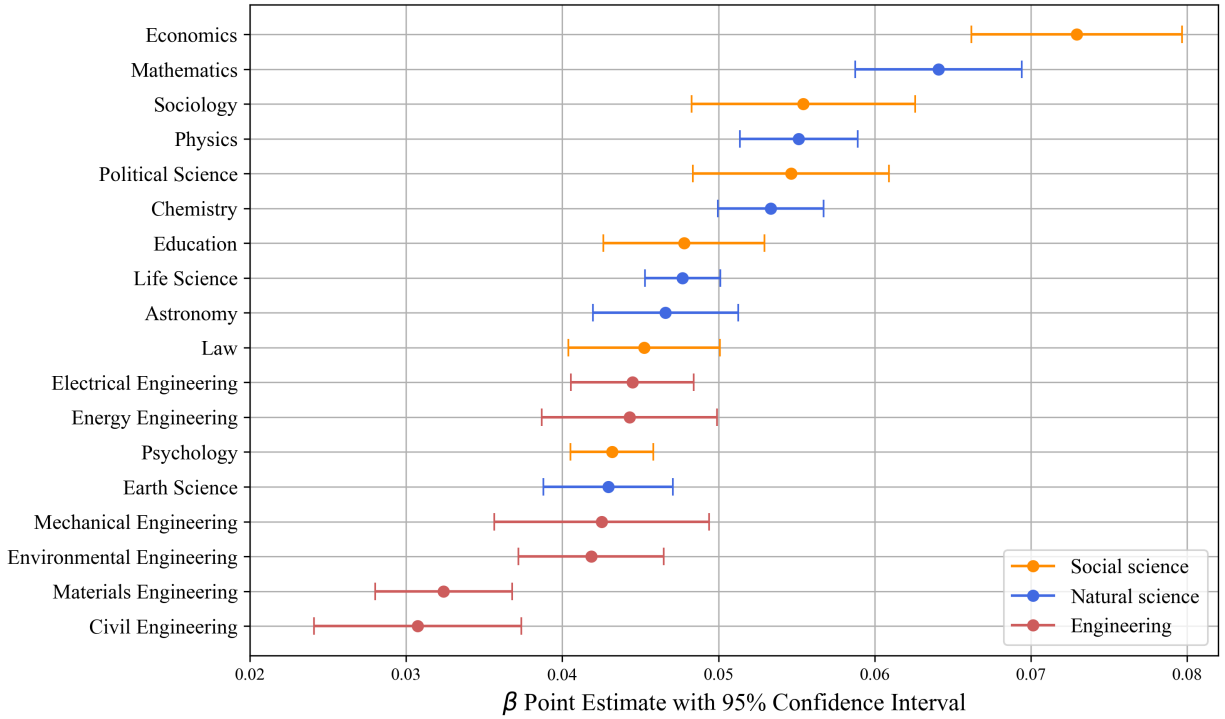
where i indexes award recipients, and t indexes years. The dependent variable y_{it} equals one if an individual i changes employing institutions in year t , and zero otherwise. The indicators $\mathbf{1}\{Field_i = f\}$ denote the recipient’s field, with each β_f giving that field’s own mobility rate. The term $\phi_{e(i,t)}$ is a set of years-since-PhD fixed effects, one per year of career age, and ψ_t a set of calendar-year fixed effects.

Figure 5 plots the average annual moving probability for each field, with 95% confidence intervals. Consistent with the claim that moving across institutions is more costly when research depends on specialized equipment and facilities, mobility tends to be lower in fields that rely more heavily on physical resources.²⁵ Economics has the highest overall mobility rate. This follows a distinct time pattern in which the high rate is driven by moves between

²⁵Mathematics is the exception among the natural sciences: it requires no specialized equipment or facilities, and it ranks second in average annual moving probability, just below economics.

the earliest post-PhD years and the award. In the first five years after the PhD, economics is not the most mobile field; fields in which postdoctoral positions are more common exhibit higher early-career mobility (Appendix Figure OA8). Economics has the highest mobility before the award, which then declines toward the levels observed in other fields (Appendix Figure OA9). Because postdoctoral appointments are less common in economics, this window corresponds roughly to the years after tenure. Over half of these pre-award, post-tenure switches are to higher-ranked institutions (Appendix Table OA11), suggesting they reflect movement toward better positions.

Figure 5: Average annual moving probabilities of award-winning researchers by field

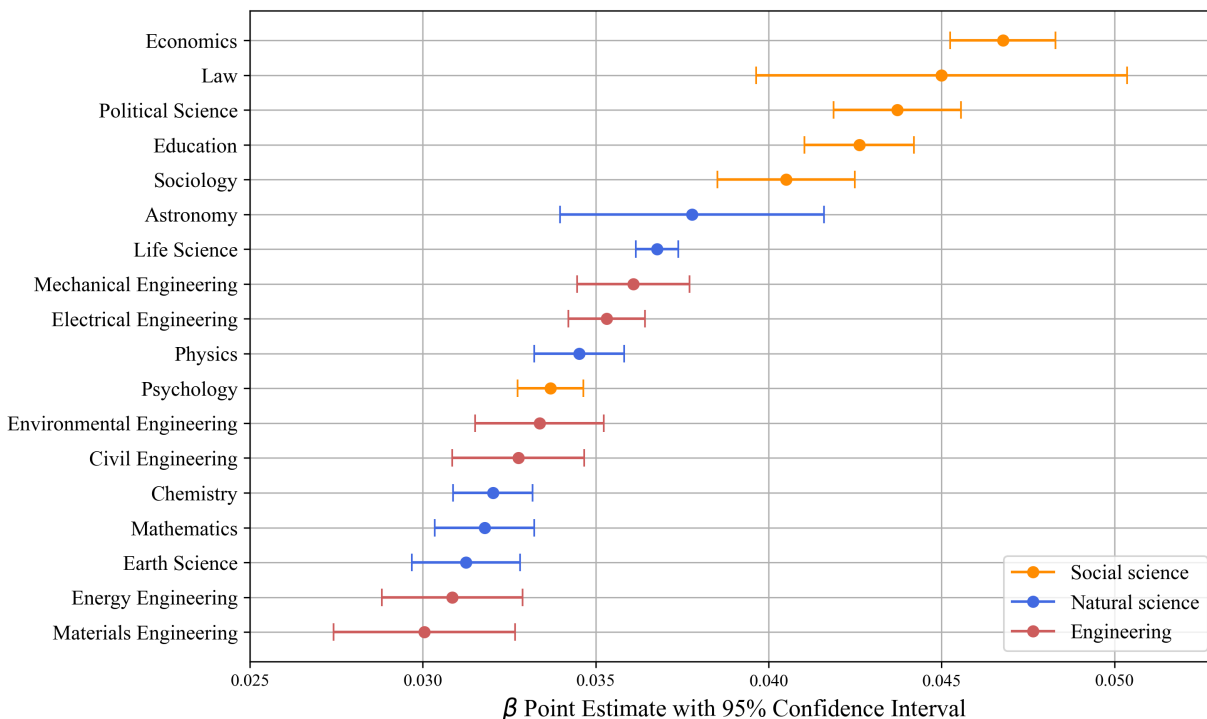


Notes: Each point is a field's average annual probability of switching institutions, with its 95% confidence interval, from a linear probability model with years-since-PhD and calendar-year fixed effects. The sample is 198,224 person-years for 4,983 award-winning researchers over 1950–2022. Standard errors are clustered by individual. Colors denote social science (orange), natural science (blue), and engineering (red).

We extend the mobility analysis to the full academic labor market and ask whether economics is still the most mobile field. Using Revelio Labs career records, we estimate equation (3) for every US PhD holder over 1990–2024. The outcome is the annual probability

that an academic employed by a university moves to a different university.²⁶ Figure 6 reports each field’s annual moving probability, adjusted for career age and calendar year. Economics is again the most mobile field, at 4.7 percent a year against a sample average of 3.6, and the fields whose research requires no dedicated equipment move 0.8 percentage points more often than the rest.

Figure 6: Average annual moving probabilities of all US PhD holders by field



Notes: Each point is a field’s average annual probability of switching institutions, with its 95% confidence interval, from a linear probability model with years-since-PhD and calendar-year fixed effects. The sample is US PhD holders employed by universities over 1990–2024, excluding the first five years after the PhD, with 1,961,286 person-years for 198,773 individuals. Standard errors are clustered by individual. Colors denote social science (orange), natural science (blue), and engineering (red).

The finding that economics is the most mobile field in both our prizewinner sample and the academic market as a whole supports the high concentration of award-winners. We use mobility as a proxy for how portable research capital is. Where physical capital is not binding, a single institution can support a larger share of a field’s leading researchers than

²⁶We count only moves between universities so that the same set of destinations is available in every field. This excludes medical centers and hospitals in the life sciences, national laboratories in physics, materials science, astronomy and energy engineering, and policy employers such as the Federal Reserve in economics. We also exclude the first five years after the PhD, which removes most postdoctoral appointments because their turnover does not reflect the cost of relocating a laboratory—the margin we want to measure.

when research is tied to specific facilities, which may contribute to the high concentration of prizewinning economists.

2.3 Nature of evidence

The final and potentially most difficult factor to quantify that contributes to the anomaly of high institutional concentration of prizewinners in economics is the nature of the evidence required to establish prizewinning research results. Our hypothesis is that because findings in natural science, engineering, and mathematics can be validated through experimental measurements or mathematical proofs, it should matter less whether the scientist is affiliated with a top-tier university when it comes to prizewinning than it does in economics.

Examples of prizewinners in mathematics and natural sciences who worked outside the major centers of science or struggled in their careers are consistent with awards going to “outsiders” for validated findings irrespective of their standing in their field. Mathematician Yitang Zhang spent years outside academia, working as an accountant and in a sandwich shop, before taking a non-tenure-track lectureship at the University of New Hampshire. When he made a major advance on the twin prime conjecture in 2013, he won a MacArthur “genius” fellowship and four other awards by 2014.²⁷ Barry Marshall and Robin Warren were little-known Australian doctors with no university backing or fame when they provided evidence that bacteria caused stomach ulcers in the 1980s; they were eventually awarded a Nobel Prize once experimental validation supported their result.²⁸ Katalin Kariko experienced repeated career setbacks in the 1980s and 1990s, including demotion and lack of funding, before her work with Drew Weissman on mRNA therapy received broad recognition and a Nobel Prize in 2023. In each case, the prizes came for validated findings irrespective of the researcher’s standing.

We test the hypothesis that economics differs from these and related fields by comparing how often an award recognizes a single testable finding rather than a broad body of work. We classify prizewinning discoveries into two categories based on the content of the recognized research and the language of the award citation. A prize is classified as recognizing a broad body of work when the cited contribution reflects a sustained research agenda developed across multiple papers over time. In contrast, a prize is classified as recognizing a single testable finding when it is based on one or a small number of papers and recognizes a

²⁷Ostrowski Prize (2013); Morningside Special Achievement Award in Mathematics (2013); Frank Nelson Cole Prize in Number Theory (2014); Rolf Schock Prize in Mathematics (2014). With the MacArthur fellowship, these awards carried roughly \$950,000 in prize money.

²⁸Barry Marshall proved that peptic ulcers are caused by bacteria by drinking a dose of the bacteria with roast beef soup. Within five days, he was vomiting and underwent an endoscopy.

specific proof, invention, discovery, or methodological contribution.²⁹ We conducted our classification with the aid of ChatGPT-4o. Appendix D gives the classification criteria, the prompt we put to the model, and worked examples. We restrict our sample to the Nobel Prizes, whose award citations take a standardized form across fields. Citations elsewhere vary by awarding organization, so classification would partly reflect drafting style.

Table 3 summarizes the classification of Nobel laureates along two dimensions: whether the prize recognizes a broad body of work or a single testable finding, and whether the laureate was affiliated with a top-ranking institution or another institution.³⁰

Table 3: 2×2 classification of selected subjects

Field (Prize)	Broad body of work		Single testable finding	
	Top 10	Other	Top 10	Other
Economics (Nobel)	36 (39%)	15 (16%)	18 (20%)	23 (25%)
Physics (Nobel)	10 (8%)	17 (13%)	28 (21%)	78 (59%)
Chemistry (Nobel)	13 (11%)	33 (29%)	12 (10%)	57 (50%)
Medicine & Physiology (Nobel)	4 (3%)	4 (3%)	30 (24%)	89 (70%)

Notes: Prizes awarded from 1969 to 2022 are included to align with the Nobel Prize in Economics. The top 10 affiliations for each subject are ranked by the number of recipients ever affiliated. See Appendix Table OA2 for the full list of each field. Classification is based on (1) the content of the prizewinning work and (2) the phrasing of the prize citation; see Appendix D for the procedure.

The results show that economics has the highest share of prizewinners from top-ranking institutions who are recognized for a broad body of work. In economics, 55 percent of Nobel laureates received the prize for a broad body of work, while 39 percent of the laureates were affiliated with top 10 institutions at the time of the award, the highest share among all fields. By contrast, medicine or physiology has the lowest shares, with 6 percent recognized for a broad body of work and 3 percent affiliated with top 10 institutions. In the other sciences, most prizewinners come from outside the top 10 and are recognized for a single testable finding, including 59 percent in physics, 50 percent in chemistry, and 70 percent in medicine.

²⁹A few examples help clarify the distinction. In economics, Paul Samuelson’s Nobel citation was “for the scientific work through which he has developed static and dynamic economic theory and actively contributed to raising the level of analysis in economic science.” In contrast, Robert C. Merton and Myron Scholes are examples of a single testable finding in economics, recognized “for a new method to determine the value of derivatives.” In chemistry, Nobel laureates Derek Barton and Odd Hassel are examples of a broad body of work, awarded “for their contributions to the development of the concept of conformation and its application in chemistry.” In contrast, Osamu Shimomura, Martin Chalfie, and Roger Y. Tsien are examples of a single testable finding, awarded “for the discovery and development of the green fluorescent protein, GFP.”

³⁰We define a top-ranking institution as one of the top 10 institutions in a given field, ranked by the number of prizewinners ever affiliated with it. The top 10 cutoff is used to provide a clear benchmark; see Appendix Table OA2 for the full list of Top 10 institutions by field.

This comparison supports the hypothesis that the type of evidence required for recognition may shape patterns of academic concentration. The high share of awards for single testable findings in the natural sciences means that results can be validated independently, so recognition depends less on being at a top university. By contrast, where awards recognize a broad body of work, as in economics, recognition depends on how the field assesses a scholar’s contributions over time rather than on a single decisive result,³¹ and institutional affiliation is one of the signals that assessment draws on.

The same mechanism applies within economics. Although the field places more weight on a broad body of work, Nobel Prizes awarded to scholars outside the most highly ranked institutions often recognize work in theory, econometrics, and analytical methods – areas in which a single principle or technique can be verified. 56 percent of awards for a single testable finding went to laureates outside the top 10, and those awards are 61 percent of all awards outside the top 10.³² These cases suggest that in economics, recognition can extend beyond the most highly ranked institutions for theoretical or methodological work.

2.4 How economics differs from the other social sciences

Economics differs not only from the natural sciences and engineering, but also from other social sciences such as sociology and political science, which fit the norm of declining institutional concentration of awards as research has spread globally. We close this section by asking what else divides economics from the other social sciences, relying in part on the comparative account of [Fourcade et al. \(2015\)](#).

We hypothesize that two related mechanisms operated to produce high and rising institutional concentration of award-winners in economics compared to the other social sciences. The first is a stronger field paradigm: economics is organized around a common core of theory and a shared set of empirical methods. The second is a greater reliance on publication in top journals that largely adhere to the field paradigm. The other social sciences also

³¹A notable example is Edward H. Clarke, whose doctoral dissertation at the University of Chicago introduced the first incentive-compatible mechanism for preference revelation, a problem considered unsolvable by many prominent economists, including Nobel laureates Paul Samuelson and George Stigler. Clarke’s committee was unconvinced that he had solved the problem and refused to award him the doctorate. Unable to support his family while developing a new dissertation, Clarke left Chicago without the degree. The University of Chicago awarded him the PhD roughly a decade later, acknowledging that his original work had indeed solved the problem ([Brannon, 2013](#)). Clarke’s mechanism later became a cornerstone of the research program recognized by the 1996 Nobel Prize in Economics, awarded to William Vickrey of Columbia. Clarke’s work was noted in the Nobel award but he was not named as an award-winner.

³²For example, Al Roth conducted his influential work in market design at the University of Pittsburgh, while William Sharpe developed the capital asset pricing model at the University of Washington. Econometricians Trygve Haavelmo (Oslo), Thomas Sargent (Minnesota), Christopher Sims (Minnesota), Robert Engle (UCSD), and Clive Granger (UCSD) also made important contributions across a range of institutional affiliations.

award prizes for scholarly books and conference volumes, and thus for research that does not readily fit into journal articles. Together, these differences create a broader space for research findings outside the major universities to gain recognition for their contributions to the field. We consider each mechanism in turn and provide evidence on the differences between economics and the other social sciences.

Field paradigm

Economics differs from the other social sciences in the greater extent to which it shares a common framework for analyzing behavior. That framework takes, at least as a first step, the assumption that workers, firms, and government regulators seek to optimize reasonably well-defined maximands in a market setting, where the key variables are prices and quantities in goods markets, and wages and employment in labor markets. The paradigm directs researchers to look first, and in some cases only, at behavior that fits into this structure (Akerlof, 2020). In addition, economics has developed econometrics as its own brand of statistics with common standards for evaluating evidence about the prices, quantities, wages, and employment (Angrist and Pischke, 2010).³³ This methodological homogeneity creates more uniform criteria for assessing research than in the other social sciences. Evidence from interdisciplinary fellowship panels further suggests that economists apply more uniform criteria in evaluating research and reach greater agreement in their assessments than scholars in other social sciences, where evaluative standards are more varied (Lamont, 2009).

Consistent with its more unified paradigm, economics has a much greater preference for within-field collaboration and a higher rate of self-citation than other social sciences. During 2000–2009, *American Political Science Review* cited the top 25 economics journals five times more frequently than *American Economic Review* cited leading political science journals. Between *American Sociological Review* and economics, the disparity is approximately eightfold (Fourcade et al., 2015). Citation data from 1995–1997 across 42 journals show that economics received 43.29% of its citations from other disciplines while citing outside fields only 9.57% of the time (Pieters and Baumgartner, 2002). In a survey of professors at American universities, only 42.1% of economists agree that “interdisciplinary knowledge is better than knowledge obtained within a single discipline”, compared with 72.9% in sociology, 59.8% in political science, and 78.7% in psychology (Fourcade et al., 2015; Gross and Simmons, 2014).

³³In a survey of graduate students from top economics departments, 78 percent of respondents agree that “economics is the most scientific of the social sciences” (Colander, 2005). The US public does not see it that way. In nationally representative surveys of US adults, Gauchat and Andrews (2018) find that 52 percent call economics “very” or “pretty” scientific and 50 percent say the same of sociology, against 91 percent for physics and 98 percent for medicine.

Recognition methods

In many social science disciplines, recognition extends beyond journal publications to a variety of discipline-specific outputs. For instance, sociology and political science place considerable weight on books (Clemens et al., 1995), with major prizes such as the Viviana Zelizer Best Book Award and the ISA Book of the Decade Award reflecting their importance as publication venues. Other disciplines rely on additional or different forms of recognition, such as practice-based contributions in law or clinical credentials in psychology (Biglan, 1973). Because recognition draws from such varied sources, researchers at institutions without top-tier journal access can still build reputations through discipline-appropriate alternative outputs.

Economics differs in its heavy reliance on journals. While books and book chapters play a role, top-tier journals are valued much more highly (Hamermesh, 2013). Publication in the “Top Five” journals has become a professional standard (Heckman and Moktan, 2020; Card and DellaVigna, 2013), as citation data show that these five journals, out of 433 in total, account for 28.5 percent of all citations and include 71 of the 100 most cited articles in the Web of Science dataset (Glötzl and Aigner, 2019).³⁴

Institutional concentration in economics is reinforced by leading journals edited at top universities (Card and DellaVigna, 2020). Wu (2007) found that between 2000 and 2003, University of Chicago affiliates authored 13.9 percent of *Journal of Political Economy* (*JPE*) articles, while Harvard and MIT authors accounted for 15.3 percent and 12.8 percent of *Quarterly Journal of Economics* (*QJE*) articles, respectively. Between 2000 and 2016, Heckman and Moktan (2020) found higher shares: 14.3 percent for Chicago in *JPE*, and 24.7 and 13.9 percent for Harvard and MIT in *QJE*. There are many plausible explanations for home bias in economics journals, including higher submission rates from local faculty and graduate students, editorial encouragement through interpersonal connections, and self-selection driven by journal style (Fourcade et al., 2015; Laband and Piette, 1994); however, similar processes operate in other fields without producing comparable effects. Other fields also host journals in university departments, such as the *American Journal of Sociology* at the University of Chicago; yet in-house authors account for only 3.7 percent of publications over the same

³⁴Glötzl and Aigner (2019) shows that, despite the growth of the economics discipline—with more authors, more institutions, and more journals—the influence of a small set of top journals and institutions has remained stable. The number of journals in the dataset rose from 35 in 1956 to 384 in 2016, yet the share of “Top Five” articles among the 100 most cited each year has stayed consistent. The top five share of total citations is due to their smaller share of overall articles, but when adjusted for this, their share of citations has increased over the past 30 years. Similarly, while the number of institutions grew to 4,337, the top twenty institutions lost only 5 percent of their share of citations in the past 25 years. Accounting for article growth, their share of the 1,000 most cited articles annually increased from 60 to 70 percent.

2000–2016 period.³⁵

3 Costs and benefits of concentration

How does the concentration of prizes among a small number of institutions affect research in economics? Concentration is, by definition, a distributional outcome, with a small number of institutions holding a large share of a field’s recognition. But concentration may also have intellectual consequences, shaping the research a field produces and making it more homogeneous in the questions it asks, in the methods it uses, or in both, than research in fields where prizes are more widely spread.

To measure the homogeneity of the methods and questions in economics compared to other social sciences, we examine the text of published abstracts. Using OpenAlex (Priem et al., 2022), we assembled 3.93 million abstracts published between 2015 and 2024 in the leading journals of six social science fields.³⁶ Every word in the shared cross-field dictionary is then tagged as describing a research method, a subject, or neither.³⁷

For each field, we draw 2,000 abstracts at random and compute the mean pairwise cosine similarity between them, separately for method words and for subject-matter words; higher values mean a field’s papers are more alike. Similarity over method words gives *method cohesion*, or how uniformly a field’s papers use the same techniques. Similarity over subject-matter words gives *topic cohesion*, or how narrowly its papers concentrate on the same questions. Because one rule is applied everywhere, the comparison does not rest on any judgment about which methods or topics belong to a particular discipline.

Figure 7 reports both measures. Economics has the highest method cohesion of the six fields (panel a): its papers use the same techniques more uniformly than those of any other social science. It has the lowest topic cohesion (panel b): its papers cover a wider range of questions than those of any other field. Appendix Figure OA10 plots the two measures against each other.

³⁵This statistic is computed from OpenAlex, which likely undercounts affiliations for this period due to incomplete metadata coverage. As a robustness check, OpenAlex yields 6.5 percent for Chicago authors in *JPE* and 10.5 and 8.5 percent for Harvard and MIT authors in *QJE* over 2000–2016. This is roughly half of the hand-coded figures of Heckman and Moktan (2020), suggesting a systematic but proportional underestimate that does not alter the relative pattern.

³⁶We restrict the comparison to the social sciences, whose research questions are broadly comparable. Papers in the natural sciences and engineering follow different writing conventions and address different kinds of questions, so their cohesion values would not be comparable to those for economics.

³⁷The dictionary contains 197,442 words. Tagging was performed by a large language model applied to the dictionary as a whole rather than field by field, so an identical rule classifies a word wherever it appears, and no field’s vocabulary is judged on its own terms. We keep substantive technique words such as *model*, *estimate*, *regression*, *survey*, and *experiment*, and drop sixteen generic research-process words such as *analysis*, *design*, *theory*, and *empirical*.

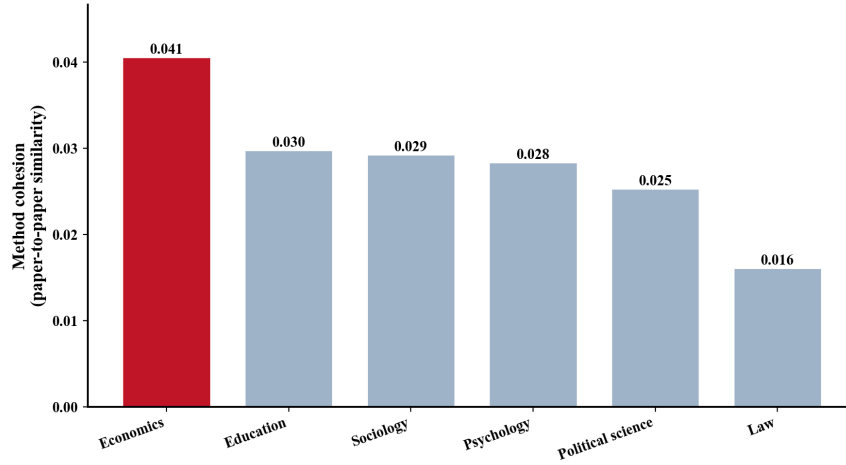
From one perspective, a shared methodology is a scientific benefit. When economists working on different subjects use the same techniques, their results can be compared and built upon, allowing readers to judge work well outside their own area. On this reading, the concentration documented above is part of how that standard is set: the institutions producing most of the field's recognized work also establish the toolkit that the rest of the field adopts. Because that toolkit travels, economics has extended beyond its traditional concerns with markets and prices into crime, family structure, discrimination, and institutions, and now ranges more widely in subject than any other social science.

But there is an alternative, less favorable view. If the institutions producing most of the field's recognized work also set its methodological standard, then meeting that standard becomes the condition of entry. The institutions that set the standard therefore shape what the field produces. On this view, work using an unfamiliar approach is hard to place or recognize, and findings that disagree with the main tenets of the field are rejected rather than studied as special cases that may yield new knowledge—winsorized away as if they have nothing to teach us. The wide range of topics is consistent with this reading as well: when the method is shared, novelty comes from the subject rather than from the approach.

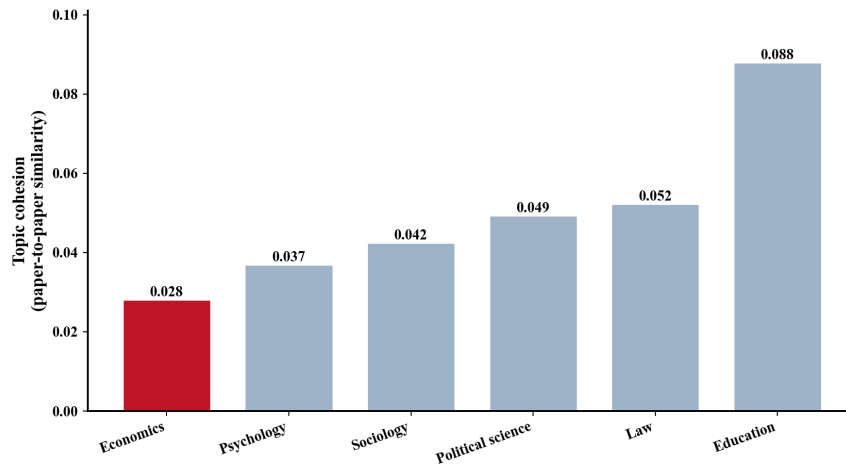
The two readings are hard to separate because they agree about the published record and disagree about the unpublished one. Both expect the papers we observe to be methodologically uniform. They differ over the work that could have appeared but did not, and over the work that was published and ignored when it could have helped us improve economic well-being—whether such analysis was set aside because its methods were inadequate to the question, or because its issues and methods were unfamiliar to the people judging it. The published record cannot decide between them.

Absent further evidence, it is reasonable to expect that both effects are at work: shared methodology has pluses and minuses, with some ideal point where marginal benefit equals marginal cost. Has the rising concentration of awards in economics gone beyond the point where the shared methodology brings the best advance in knowledge, has it reached that point, or will more concentration produce more gains in knowledge in economics? The divergence of economics from all other fields, including social sciences, natural sciences, engineering, and mathematics, makes it hard to dismiss the interpretation that the costs have moved beyond the benefits. On either reading, the data establish that economics has moved against a trend in every other discipline we examine and that the divergence has widened rather than narrowed.

Figure 7: Method and topic cohesion across the social sciences



(1) Method cohesion



(2) Topic cohesion

Notes: Each bar reports the mean pairwise cosine similarity between papers within a field, computed from a random sample of 2,000 abstracts per field drawn from 3.93 million abstracts published in each field’s leading journals, 2015–2024. Every word in the cross-field vocabulary (197,442 terms) is classified as describing either research methods or subject matter, using a single classification rule applied uniformly to all fields. Panel (a) measures similarity over method words, capturing how uniformly a field’s papers employ the same techniques; panel (b) measures similarity over subject-matter words, capturing how narrowly a field’s papers concentrate on the same topics. Higher values indicate greater within-field homogeneity. Economics (red) displays the highest method cohesion and the lowest topic cohesion of the six fields: its papers are the most alike in how they conduct research and the most varied in what they study.

4 Conclusion

In this paper, we collected lifetime biographical data on award-winning scientists in the expectation that as science spreads globally and within countries, the affiliations of award-winners would move beyond a few centers in advanced, wealthy countries to other institutions in those countries and to the rest of the world. The data support this hypothesis in seventeen major disciplines across the natural sciences, engineering, and the social sciences with the glaring exception of economics where the concentration of awards increased over time to make it the most institutionally concentrated discipline of all.

Analyzing the anomaly, we identified four features of economics associated with the high and rising institutional concentration of awards: (1) an allocation of talent in which prizewinners are affiliated with top-ranked institutions over their entire career more than prizewinners in any other field; (2) the absence of expensive laboratories or equipment that tie work to physical capital, which allows economists to move between institutions more often than researchers in other fields and lets a few departments attract an exceptional share of star researchers; (3) a production of knowledge based on expert agreement more than experimental or mathematical proof; and (4) a more cohesive paradigm than in other social sciences with narrower channels of recognition and methodology that makes affiliation with the institutions holding that standard more consequential in careers.

While we focused on the anomaly, the fact that all of the other sciences, including the other social sciences, fit with the expectation that worldwide expansion of science has been accompanied by wider distribution of prizes of excellence is worth attention and further study in its own right as the general pattern in the development of science, as well as for the insight it may cast on the anomaly.

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Online Appendix

A Data summary statistics

Table OA1: List of awards

Award name and website	# recipients
Mathematics	
The Abel Prize	25
Fields Medal	64
Wolf Prize in Mathematics	65
Crafoord Prize in Mathematics	13
The Shaw Prize in Mathematical Sciences	29
Rolf Nevanlinna Prize	11
The Mirzakhani Prize—formerly the NAS Award in Mathematics	9
Bocher Memorial Prize	37
George David Birkhoff Prize in Applied Mathematics	19
Norbert Wiener Prize in Applied Mathematics	20
Oswald Veblen Prize in Geometry	37
Leroy P. Steele Prize for Lifetime Achievement	33
Rolf Schock Prize in Mathematics	14
Breakthrough Prize in Mathematics	14
Physics	
Nobel Prize in Physics	222
Wolf Prize in Physics	68
Isaac Newton Medal	15
Max Planck Medal	86
Breakthrough Prize in Fundamental Physics	38
Special Breakthrough Prize in Fundamental Physics	16
Physics Frontiers Prize	10
Lorentz Medal	24
Henri Poincare Prize	30
Benjamin Franklin Medal in Physics	39
UNESCO Niels Bohr Medal	10
Chemistry	
Continued on next page	

Table OA1: List of awards (continued)

Award name and website	# recipients
Nobel Prize in Chemistry	191
Wolf Prize in Chemistry	61
Priestley Medal	88
Welch Award in Chemistry	56
NAS Award in Chemical Science	43
Benjamin Franklin Medal in Chemistry	26
Faraday Lectureship Prize	37
The Davy Medal	146
Peter Debye Award for Physical Chemistry	56
Roger Adams Award in Organic Chemistry	33
Life science	
Nobel Prize in Physiology or Medicine	225
Albert Lasker Basic Medical Research Award	167
Lasker-DeBakey Clinical Medical Research Award	156
Lasker-Koshland Special Achievement Award in Medical Science	17
The Canada Gairdner International Award	367
Canada Gairdner Global Health Award	17
The Shaw Prize in Life Science and Medicine	37
Wolf Prize in Medicine	64
Crafoord Prize in Biosciences	17
The Kavli Prize in Neuroscience	24
Breakthrough Prize in Life Sciences	58
Benjamin Franklin Medal in Life Science	28
Heineken Prize for Medicine	18
Heineken Prize for Biochemistry and Biophysics	27
Astronomy	
Crafoord Prize in Astronomy	13
The Kavli Prize in Astrophysics	19
The Shaw Prize in Astronomy	36
The Gold Medal from the Royal Astronomical Society	257
The Bruce Medal	114
Dannie Heineman Prize for Astrophysics	48
Continued on next page	

Table OA1: List of awards (continued)

Award name and website	# recipients
James Craig Watson Medal	48
Henry Draper Medal	56
Earth science	
Crafoord Prize in Geosciences	17
Wollaston Medal	195
Penrose Medal	97
Vetlesen Prize	33
Benjamin Franklin Medal in Earth and Environmental Science	25
Arthur L. Day Prize and Lectureship	18
Arthur L. Day Medal	76
Carl-Gustaf Rossby Research Medal in Atmospheric Science	68
Alexander Agassiz Medal in Oceanography	48
A.G. Huntsman Award for Excellence in Marine Sciences	47
G. K. Warren Prize for Fluvial Geology	13
International Meteorological Organization Prize	67
Electrical Engineering	
Turing Award	76
IEEE Medal of Honor	103
IEEE Edison Medal	110
IEEE John von Neumann Medal	34
Benjamin Franklin Medal in Electrical Engineering	25
Benjamin Franklin Medal in Computer and Cognitive Science	27
The Okawa Prize	58
Knuth Prize	22
Royal Society Milner Award	11
W. Wallace McDowell Award	32
BBVA Foundation Frontiers of Knowledge Award in Information and Communication Technologies	21
Civil engineering	
Freyssinet Medal	13
IABSE Medal of Merit	45
IABSE Honorary Membership	67
Continued on next page	

Table OA1: List of awards (continued)

Award name and website	# recipients
Theodore von Karman Medal	61
FIB Medal of Merit	30
Energy engineering	
Eni Award	98
Enrico Fermi Award	65
The Global Energy Award	47
Environmental engineering	
Tyler Prize for Environmental Achievements	79
Volvo Environmental Prize	51
Stockholm Water Prize	34
BBVA Foundation Frontiers of Knowledge Award in Ecology and Conservation Biology	27
BBVA Foundation Frontiers of Knowledge Award in Climate Change	22
Heineken Prize in Environmental Sciences	17
The Zayed International Prize for the Environment	9
Materials engineering	
MRS Von Hippel Award	47
MRS Medal	55
David Turnbull Lectureship	32
Outstanding Early Career Investigator Award	36
Armourers and Braisiers Company Prize	20
Mechanical engineering	
ASME Medal	91
Timoshenko Medal	70
Benjamin Franklin Medal in Mechanical Engineering	13
Gibbs Brothers Medal	18
Economics	
The Sveriges Riksbank Prize in Economic Sciences	93
John Bates Clark Medal	45
The Frisch Medal Award	44
The John von Neumann Award	29
The IZA Prize in Labor Economics	21
Continued on next page	

Table OA1: List of awards (continued)

Award name and website	# recipients
The Jacob Mincer Award	22
The Deutsche Bank Prize in Financial Economics	6
Stephen A. Ross Award	16
Leontief Prize for Advancing the Frontiers of Economic Thought	34
The Erwin Plein Nemmers prize in Economics	14
The Fischer Black Prize	10
BBVA Foundation Frontiers of Knowledge Award in Economics, Finance and Management	25
Political Science	
The Johan Skytte Prize in Political Science	29
ECPR Lifetime Achievement Award	10
Karl Deutsch Award of the International Political Science Association	10
Karl Deutsch Award of the International Studies Association	39
ISA Foreign Policy Analysis Section Distinguished Scholar Award	32
ISA International Political Economy Section Distinguished Scholar Award	33
ISA THEORY Distinguished Scholar Award	7
ISA Book of the Decade Award	7
IPSA Foundation Mattei Dogan Award	7
Juan Linz Prize	4
Sir Isaiah Berlin Prize	23
Sociology	
ISA Award for Excellence in Research and Practice	3
European Amalfi Prize for Sociology and Social Sciences	21
British Journal of Sociology Prize	4
Adam Podgorecki Prize	20
The Balzan Prize	4
Roger V. Gould Prize	34
William F. Ogburn Career Achievement Award	22
Viviana Zelizer Best Book Award	33
W.E.B. Du Bois Career of Distinguished Scholarship Award	44
Linton C. Freeman Award	13
Law	
Continued on next page	

Table OA1: List of awards (continued)

Award name and website	# recipients
The Stockholm Prize in Criminology	32
Law and Society Association International Prize	18
Harry J. Kalven, Jr. Prize	46
Peter Birks Prize	10
Francis Deak Prize	52
Canada Prize of the International Academy of Comparative Law	7
Hessel Yntema Prize	31
Manley O'Hudson Medal	44
European Association of Law and Economics Award	12
European Criminology Award	16
Edwin H. Sutherland Award	62
Tang Prize in the Rule of Law	5
Education	
AERA Distinguished Contributions to Research in Education Award	62
E. F. Lindquist Award	52
E. L. Thorndike Career Achievement Award	60
AERA Outstanding Book Award	58
Oeuvre Award for Outstanding Contributions to the Science of Learning and Instruction	16
John Nisbet Award	18
Yidan Prize	15
LRA Distinguished Scholar Lifetime Achievement Award	21
Grawemeyer Award in Education	41
NCME Career Contribution Award to Educational Measurement	11
SSSR Distinguished Scientific Contributions Award	11
CIES Honorary Fellows Award	38
Psychology	
APA Award for Distinguished Scientific Contributions	217
APA Award for Outstanding Lifetime Contributions to Psychology	35
APA Award for Distinguished Contributions to the International Advancement of Psychology	38
APA Award for Distinguished Professional Contributions to	48
Continued on next page	

Table OA1: List of awards (continued)

Award name and website	# recipients
Applied Research	
APA Distinguished Scientific Award for the Applications of Psychology	56
APA Distinguished Scientific Award for an Early Career	185
Contribution to Psychology	
APS James McKeen Cattell Fellow Award	104
APS William James Fellow Award	187
APS James S. Jackson Lifetime Achievement Award for	7
Transformative Scholarship	
Atkinson Prize in Psychological and Cognitive Sciences	12

Table OA2: Top 10 affiliations for each subject ranked by the number of recipients

Institute name	# of recipients	% of recipients
Mathematics		
Number of Awards: 14		
Number of Recipients: 261		
Number of Institutions: 233		
Princeton University	87	33.33
Institute for Advanced Study	57	21.84
Harvard University	57	21.84
Massachusetts Institute of Technology	56	21.46
University of California, Berkeley	47	18.01
New York University	43	16.48
University of Chicago	40	15.33
Stanford University	39	14.94
École normale supérieure	23	8.81
University of Cambridge	22	8.43
Physics		
Number of Awards: 11		
Number of Recipients: 427		
Number of Institutions: 390		
Princeton University	67	15.69
Harvard University	59	13.82
University of Cambridge	54	12.65
University of California, Berkeley	51	11.94
Massachusetts Institute of Technology	50	11.71
University of Chicago	38	8.90
California Institute of Technology	38	8.90
Stanford University	37	8.67
CERN	37	8.67
Columbia University	35	8.20
Chemistry		
Number of Awards: 10		
Number of Recipients: 495		
Number of Institutions: 490		
Continued on next page		

Table OA2: Top 10 affiliations (continued)

Institute name	# of recipients	% of recipients
Harvard University	94	18.99
University of Cambridge	66	13.33
University of California, Berkeley	63	12.73
University of Oxford	54	10.91
Massachusetts Institute of Technology	52	10.51
Columbia University	45	9.09
California Institute of Technology	41	8.28
Imperial College London	38	7.68
University of Chicago	36	7.27
Stanford University	34	6.87
Life Science		
Number of Awards: 14		
Number of Recipients: 762		
Number of Institutions: 682		
Harvard University	140	18.37
National Institutes of Health	86	11.29
University of Cambridge	79	10.37
Johns Hopkins University	67	8.79
Rockefeller University	64	8.40
Columbia University	63	8.27
Yale University	54	7.09
University of California, Berkeley	50	6.56
University of Oxford	47	6.17
Stanford University	45	5.91
Astronomy		
Number of Awards: 8		
Number of Recipients: 428		
Number of Institutions: 257		
University of Cambridge	76	17.76
California Institute of Technology	55	12.85
Princeton University	43	10.05
Harvard University	41	9.58
Continued on next page		

Table OA2: Top 10 affiliations (continued)

Institute name	# of recipients	% of recipients
University of Königsberg	40	9.35
University of California, Berkeley	35	8.18
Ohio State University	32	7.48
University of Chicago	30	7.01
Massachusetts Institute of Technology	25	5.84
University of Oxford	22	5.14
Earth Science		
Number of Awards: 12		
Number of Recipients: 453		
Number of Institutions: 389		
Massachusetts Institute of Technology	63	13.91
University of Cambridge	57	11.04
Harvard University	50	11.04
University of Chicago	41	9.05
Columbia University	33	7.28
United States Geological Survey	32	7.06
California Institute of Technology	29	6.40
University of California, San Diego	29	6.40
Yale University	29	6.40
Princeton University	26	5.74
Electrical Engineering		
Number of Awards: 11		
Number of Recipients: 444		
Number of Institutions: 338		
Massachusetts Institute of Technology	105	23.65
Bell Laboratories	73	16.44
Stanford University	67	15.09
University of California, Berkeley	60	13.51
Harvard University	49	11.04
IBM Laboratories	38	8.56
Carnegie Mellon University	33	7.43
Princeton University	32	7.21
Continued on next page		

Table OA2: Top 10 affiliations (continued)

Institute name	# of recipients	% of recipients
Columbia University	31	6.98
University of Illinois-Urbana Champaign	26	5.86
Civil Engineering		
Number of Awards: 5		
Number of Recipients: 171		
Number of Institutions: 177		
University of California, Berkeley	17	9.94
Columbia University	13	7.60
ETH Zurich	12	7.02
California Institute of Technology	12	7.02
Brown University	12	7.02
Massachusetts Institute of Technology	11	6.43
Lehigh University	10	5.85
University of Cambridge	9	5.26
University of Illinois-Urbana Champaign	9	5.26
Northwestern	8	4.68
Energy Engineering		
Number of Awards: 3		
Number of Recipients: 196		
Number of Institutions: 281		
University of California, Berkeley	28	14.29
Massachusetts Institute of Technology	19	9.69
Stanford University	18	9.18
University of Chicago	18	9.18
California Institute of Technology	15	7.65
Harvard University	13	6.63
University of Cambridge	13	6.63
Cornell University	12	6.12
Russian Academy of Sciences	12	6.12
Princeton University	12	6.12
Environmental Engineering		
Number of Awards: 7		
Continued on next page		

Table OA2: Top 10 affiliations (continued)

Institute name	# of recipients	% of recipients
Number of Recipients: 186		
Number of Institutions: 301		
University of California, Berkeley	24	12.90
Harvard University	23	12.37
Stanford University	16	8.60
Massachusetts Institute of Technology	14	7.53
Yale University	12	6.45
Princeton University	12	6.45
University of Cambridge	12	6.45
University of California, San Diego	11	5.91
Cornell University	11	5.91
University of Oxford	11	5.91
Materials Engineering		
Number of Awards: 5		
Number of Recipients: 166		
Number of Institutions: 162		
Massachusetts Institute of Technology	33	19.88
University of Cambridge	28	16.87
University of California, Berkeley	21	12.65
Harvard University	21	12.65
University of Illinois-Urbana Champaign	18	10.84
Stanford University	18	10.84
Northwestern University	16	9.64
Princeton University	11	6.63
California Institute of Technology	10	6.02
Cornell University	10	6.02
Mechanical Engineering		
Number of Awards: 4		
Number of Recipients: 165		
Number of Institutions: 169		
Massachusetts Institute of Technology	31	18.79
Brown University	22	13.33
Continued on next page		

Table OA2: Top 10 affiliations (continued)

Institute name	# of recipients	% of recipients
University of California, Berkeley	16	9.70
University of Cambridge	15	9.09
Stanford University	15	9.09
California Institute of Technology	15	9.09
Cornell University	15	9.09
Columbia University	11	6.67
Harvard University	11	6.67
University of Illinois-Urbana Champaign	9	5.45
Economics		
Number of Awards: 12		
Number of Recipients: 259		
Number of Institutions: 249		
Harvard University	96	37.07
Massachusetts Institute of Technology	80	30.89
University of Chicago	65	25.10
Yale University	59	22.78
Princeton University	58	22.39
Stanford University	55	21.24
University of California, Berkeley	46	17.76
Columbia University	36	13.90
London School of Economics	32	12.36
University of Oxford	27	10.42
Political Science		
Number of Awards: 11		
Number of Recipients: 158		
Number of Institutions: 227		
Harvard University	38	24.05
Stanford University	30	18.99
Columbia University	26	16.46
Yale University	24	15.19
University of Chicago	20	12.66
Princeton University	19	12.03
Continued on next page		

Table OA2: Top 10 affiliations (continued)

Institute name	# of recipients	% of recipients
University of Oxford	17	10.76
University of Michigan	17	10.76
University of California, Berkeley	16	10.13
Ohio State University	13	8.23
Sociology		
Number of Awards: 10		
Number of Recipients: 185		
Number of Institutions: 264		
Columbia University	33	17.84
Harvard University	33	17.84
University of Chicago	31	16.76
University of California, Berkeley	30	16.22
Stanford University	19	10.27
University of Michigan	18	9.73
Princeton University	17	9.19
Northwestern University	14	7.57
University of Wisconsin-Madison	13	7.03
New York University	13	7.03
Law		
Number of Awards: 12		
Number of Recipients: 252		
Number of Institutions: 274		
Harvard University	47	18.65
Yale University	37	14.68
Columbia University	29	11.51
University of Oxford	25	9.92
University of Chicago	25	9.92
University of Cambridge	25	9.92
University of California, Berkeley	19	7.54
University of Michigan	15	5.95
New York University	15	5.95
University of Wisconsin-Madison	14	5.56
Continued on next page		

Table OA2: Top 10 affiliations (continued)

Institute name	# of recipients	% of recipients
Education		
Number of Awards: 12		
Number of Recipients: 294		
Number of Institutions: 335		
Harvard University	58	19.73
Stanford University	52	17.69
University of Chicago	43	14.63
Columbia University	32	10.88
University of California, Berkeley	28	9.52
University of Michigan	28	9.52
University of Illinois-Urbana Champaign	25	8.50
University of California, Los Angeles	24	8.16
University of Wisconsin-Madison	21	7.14
University of Pennsylvania	20	6.80
Psychology		
Number of Awards: 10		
Number of Recipients: 589		
Number of Institutions: 360		
Harvard University	133	22.58
Yale University	94	15.96
University of Michigan	75	12.73
Stanford University	74	12.56
University of Pennsylvania	62	10.53
University of California, Berkeley	52	8.83
Columbia University	47	7.98
University of Illinois-Urbana Champaign	44	7.47
University of California, Los Angeles	44	7.47
University of Minnesota	43	7.30

Table OA3: Ranking of economics affiliations by share of affiliation-years

Institute name	% time	Ranking
Harvard University	10.94	1
University of Chicago	8.03	2
Massachusetts Institute of Technology	7.63	3
Stanford University	6.40	4
Princeton University	4.88	5
Yale University	4.25	6
University of California, Berkeley	4.02	7
Columbia University	3.34	8
London School of Economics and Political Science	3.03	9
University of Oxford	2.32	10
University of Pennsylvania	2.17	11
University of Cambridge	2.16	12
Cornell University	1.73	13
Carnegie Mellon University	1.58	14
Northwestern University	1.53	15
New York University	1.35	16
Hebrew University of Jerusalem	1.31	17
University of Minnesota	1.25	18
University of Massachusetts Amherst	1.14	19
Tel Aviv University	1.08	20
University College London	0.95	21
University of Oslo	0.91	22
University of Michigan	0.87	23
University of California, Los Angeles	0.87	23
University of Manchester	0.77	25
Stockholm University	0.69	26
University of Sussex	0.68	27
University of Maryland-College Park	0.67	28
Boston University	0.62	29
University of California, San Diego	0.59	30
The New School	0.54	31
University of Texas at Austin	0.54	32
Continued on next page		

Table OA3: Ranking of economics affiliations by time share (continued)

Institute name	% of affiliation-years	Ranking
University of California,Santa Barbara	0.53	33
University of Wisconsin-Madison	0.52	34
Brown University	0.51	35
École Nationale Supérieure des Mines de Nancy	0.48	36
Washington University in St.Louis	0.47	37
Erasmus University Rotterdam	0.47	38
University of Washington	0.41	39
University of Delhi	0.38	40
California Institute of Technology	0.37	41
Indiana University	0.37	41
University of Southern California	0.37	43
Autonomous University of Barcelona	0.37	43
Stockholm School of Economics	0.36	45
Jawaharlal Nehru University	0.35	46
City College of New York	0.35	46
RAND Corporation	0.32	48
Institute for Advanced Study	0.29	49
Duke University	0.29	49

B Additional figures and tables

Figure OA1: Normalized HHI: Natural Sciences

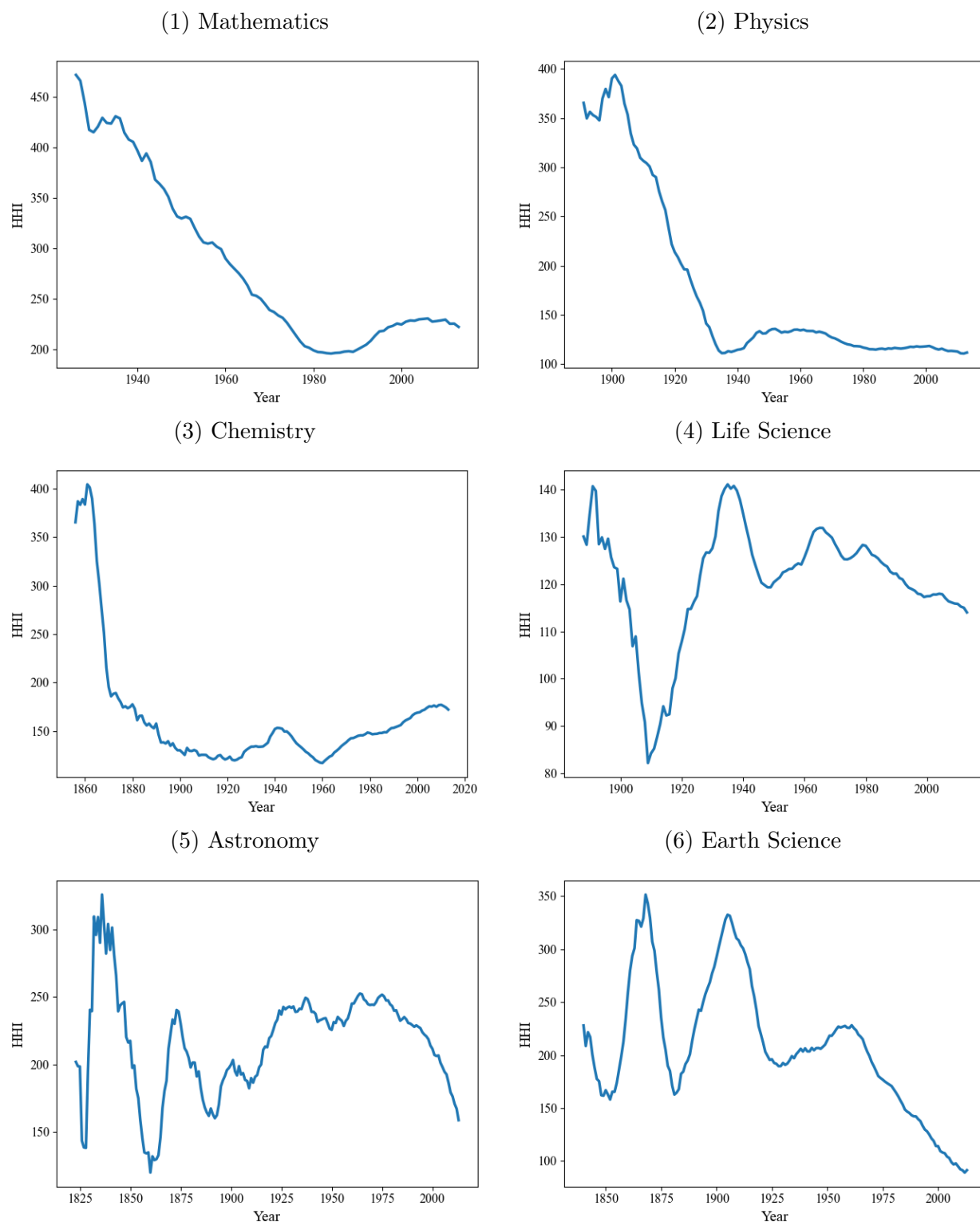


Figure OA1: Normalized HHI (continued): Engineering

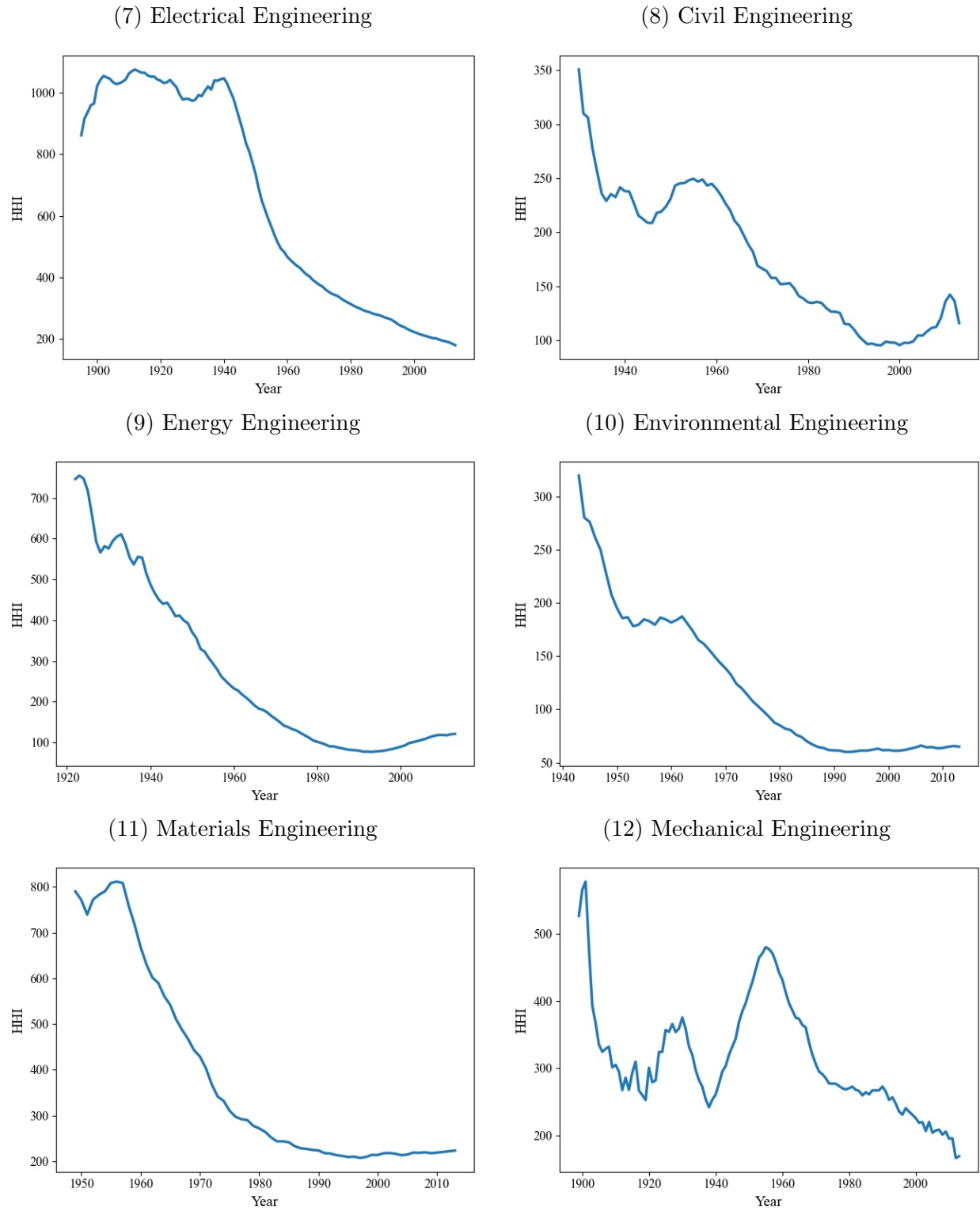
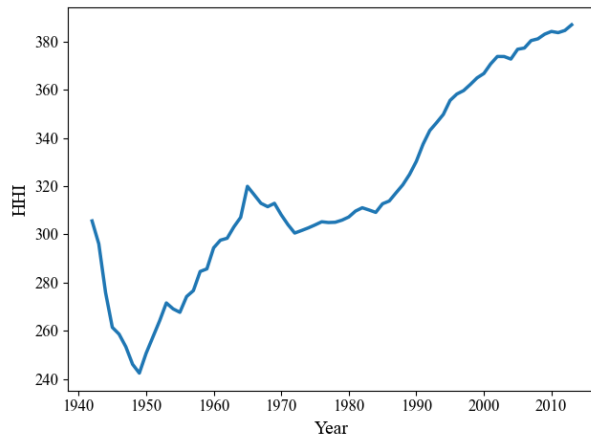
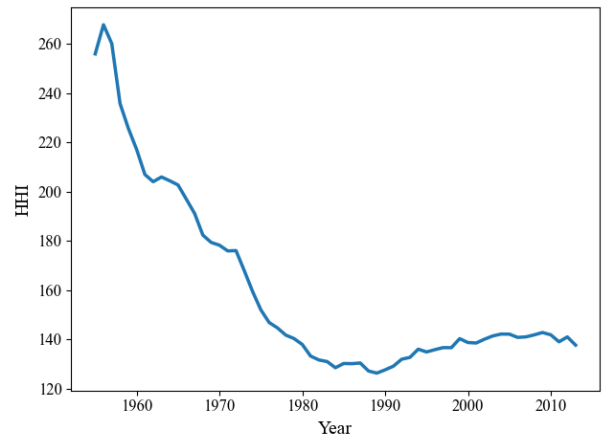


Figure OA1: Normalized HHI (continued): Social Sciences

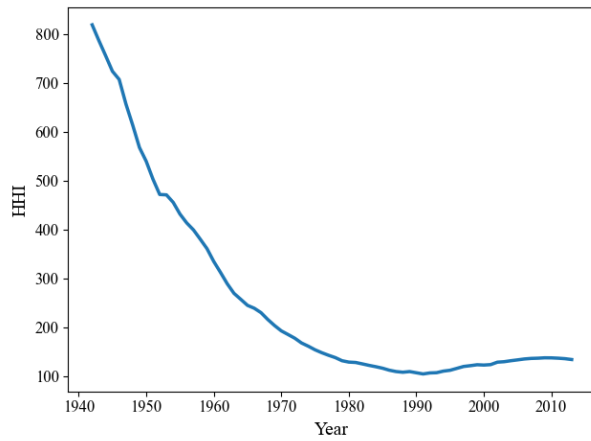
(13) Economics



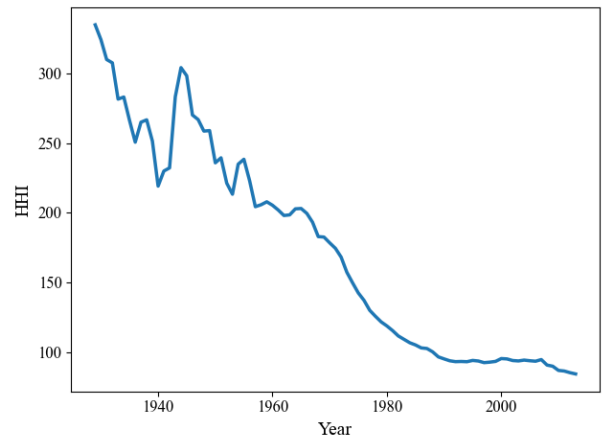
(14) Political Science



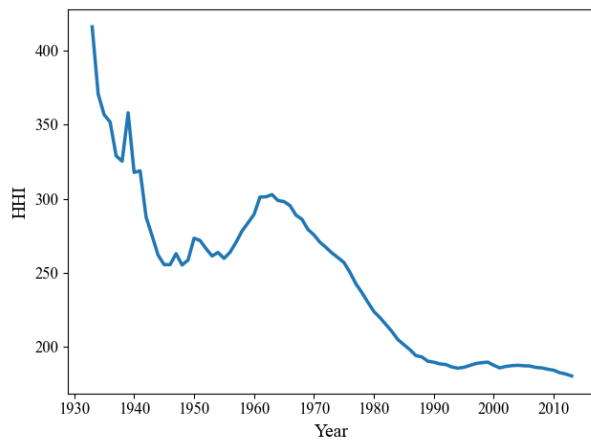
(15) Sociology



(16) Law



(17) Education



(18) Psychology

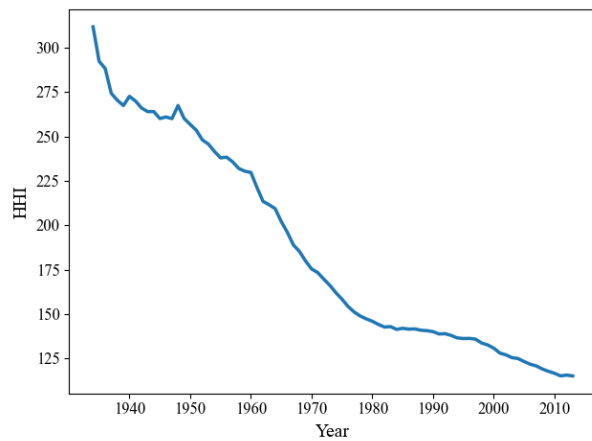


Figure OA2: Normalized employment and education HHI trends by prizewinner's affiliation

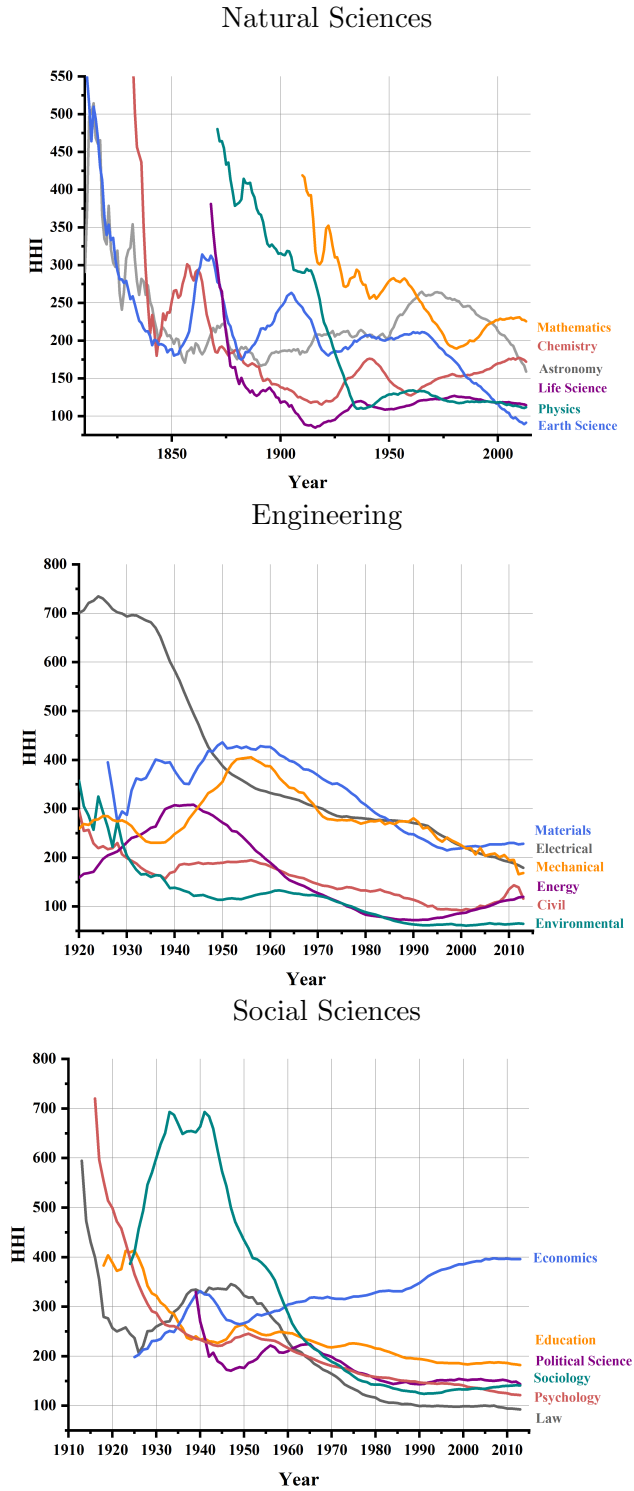


Figure OA3: Normalized employment HHI trends by prizewinner's affiliation after 1950

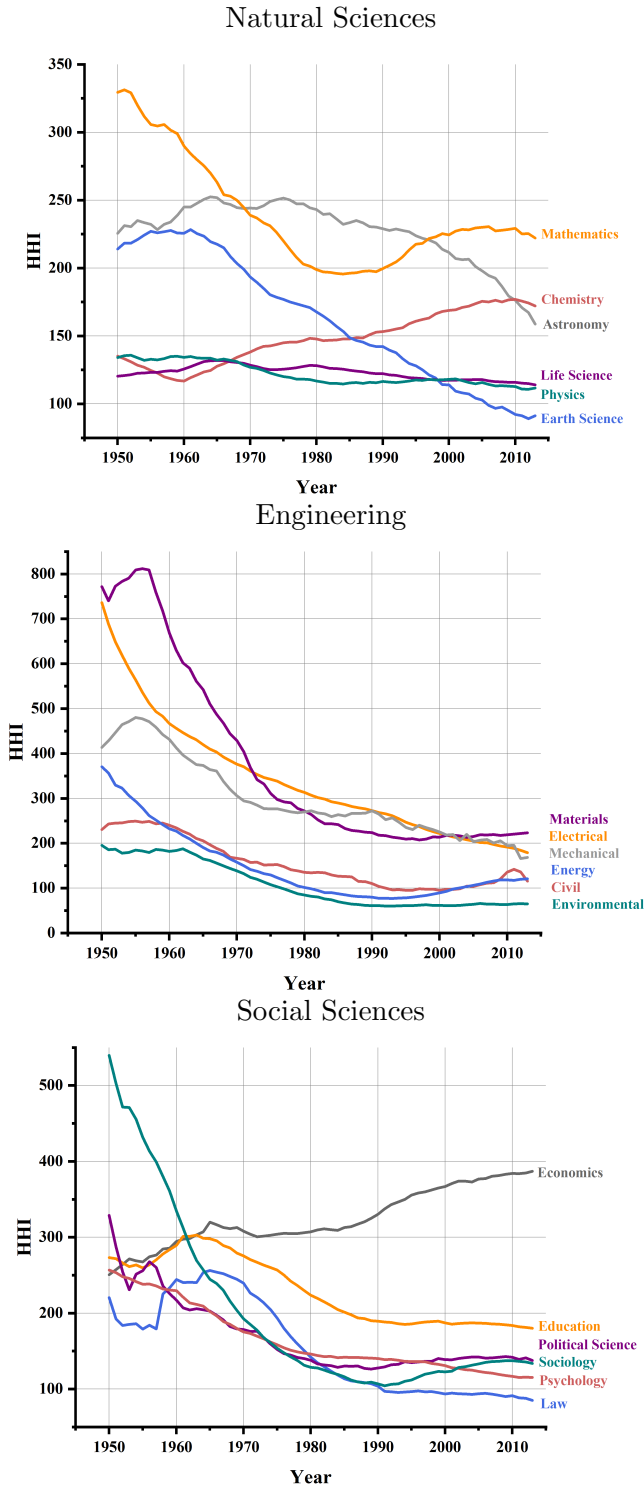


Figure OA4: Normalized employment HHI trends by prizewinner's affiliation until laureate year

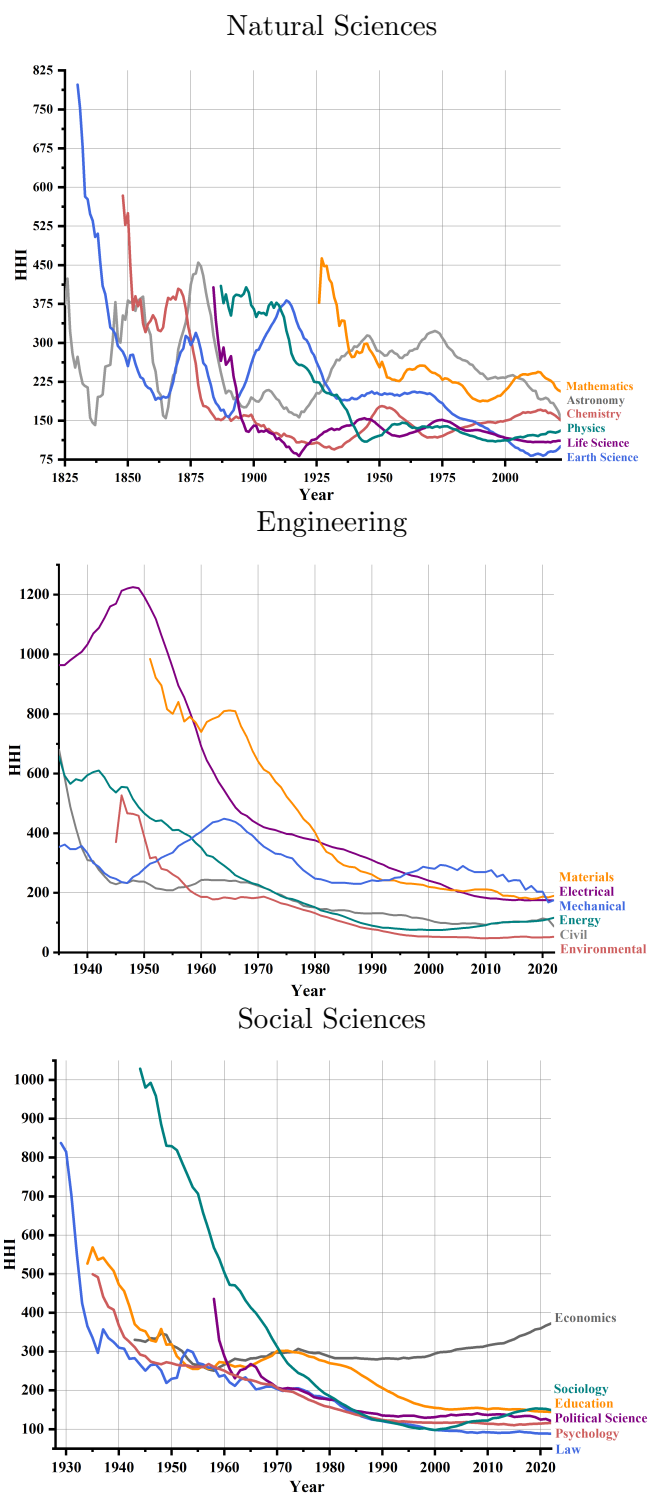
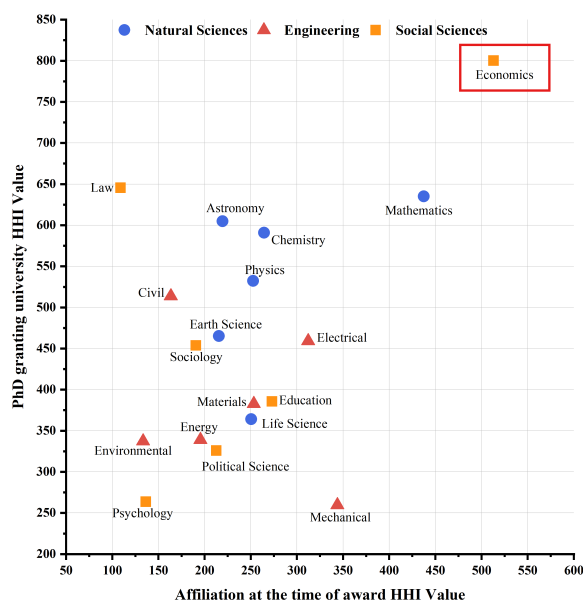
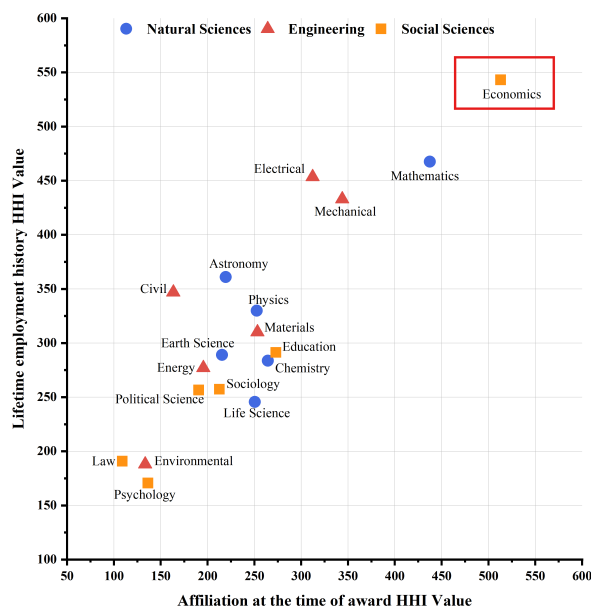


Figure OA5: Normalized HHI for affiliation of prizewinning researchers at the time of award, PhD granting university and lifetime employment history: US institutions only



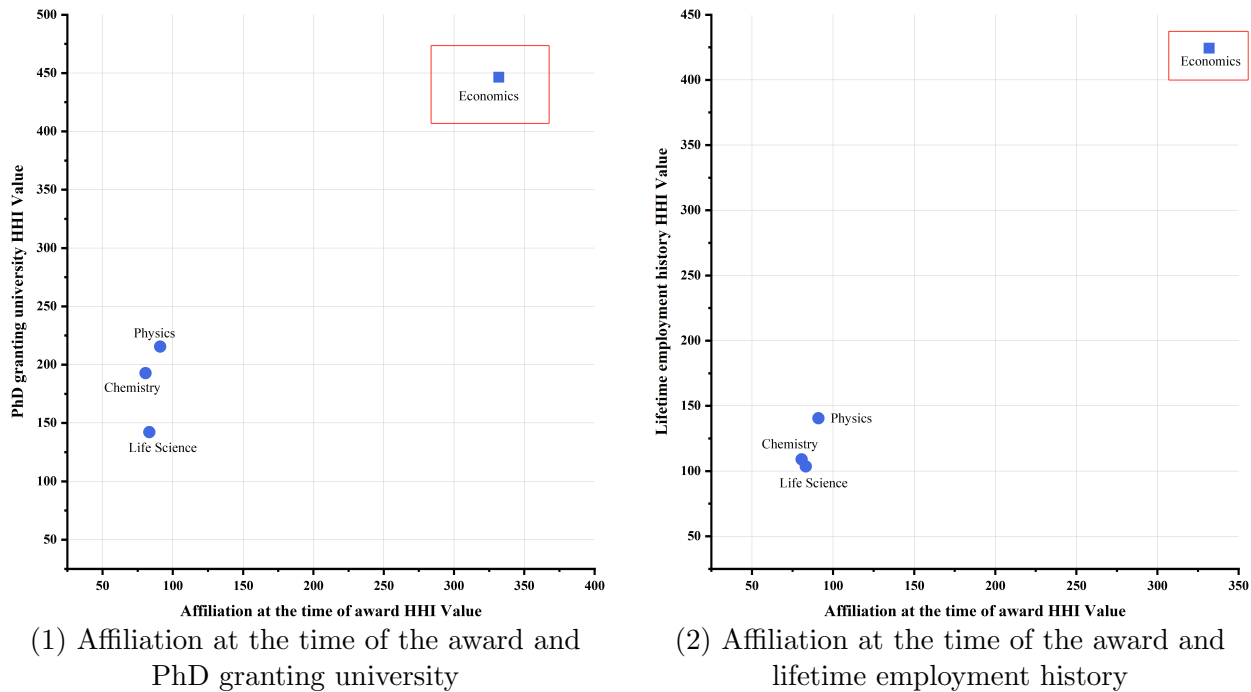
(1) Affiliation at the time of the award and PhD granting university



(2) Affiliation at the time of the award and lifetime employment history

Notes: Affiliation at the time of the award is determined based on the institution with which the individual was associated during the year the award was conferred. The PhD-granting institution is identified as the university where the individual obtained their doctoral degree. Lifetime affiliation is computed by considering all institutions with which the individual has been affiliated, with a time-weighted approach to account for the duration of each affiliation.

Figure OA6: Normalized HHI for affiliation at time of award, lifetime affiliation and PhD affiliation for Nobel prizewinners



Notes: Affiliation at the time of the award is determined based on the institution with which the individual was associated during the year the award was conferred. The PhD-granting institution is identified as the university where the individual obtained their doctoral degree. Lifetime affiliation is computed by considering all institutions with which the individual has been affiliated, with a time-weighted approach to account for the duration of each affiliation.

Table OA4: Different measures of institutional concentration of laureates

Academic Field	50% share of affiliation-years in #/% institutions	90% share of affiliation-years in #/% institutions	Gini coefficient
Natural Sciences			
Mathematics	13 (5.5%)	76 (32.8%)	0.761
Physics	26 (6.6%)	142 (36.5%)	0.731
Chemistry	27 (5.5%)	173 (35.2%)	0.750
Life Science	32 (4.6%)	236 (34.7%)	0.760
Astronomy	18 (7.1%)	96 (37.5%)	0.722
Earth Science	23 (5.8%)	152 (39.2%)	0.725
Engineering			
Electrical Engineering	12 (3.7%)	100 (29.5%)	0.793
Civil Engineering	17 (9.8%)	71 (40.3%)	0.669
Energy Engineering	29 (10.5%)	133 (47.2%)	0.635
Environmental Engineering	32 (10.6%)	134 (44.5%)	0.645
Materials Engineering	10 (6.1%)	63 (38.9%)	0.715
Mechanical Engineering	11 (6.6%)	65 (38.5%)	0.719
Social Sciences			
Economics	8 (3.3%)	67 (27.0%)	0.812
Political Science	18 (8.1%)	94 (41.2%)	0.693
Sociology	19 (7.0%)	107 (40.5%)	0.707
Law	23 (8.5%)	116 (42.4%)	0.682
Education	17 (5.1%)	118 (35.3%)	0.746
Psychology	18 (5.1%)	101 (28.2%)	0.784

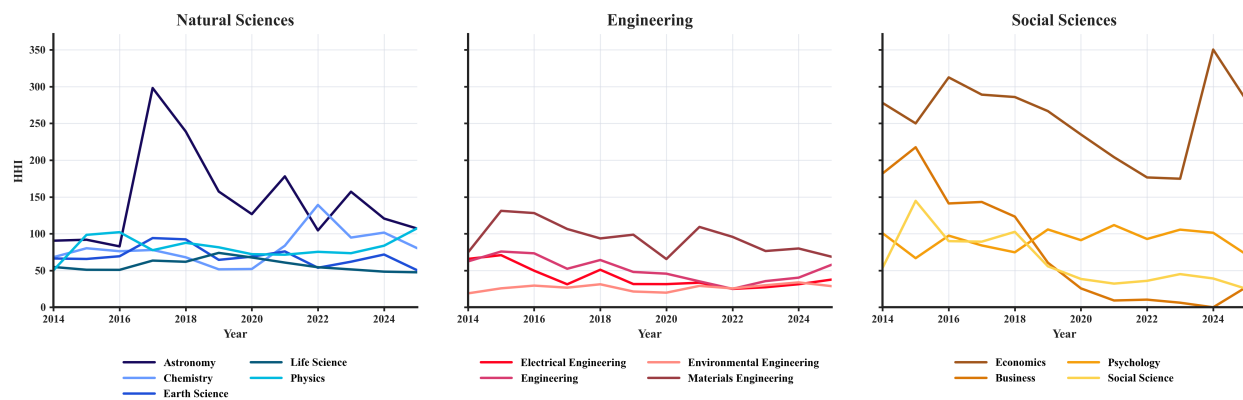
Notes: This table presents results from different measures of concentration for the 18 fields: (1) Number or percentage of institutions accounting for a 50% share of laureates' career time; (2) Number or percentage of institutions contributing to a 90% share of laureates' career time; and (3) Gini coefficient reflecting the distribution of laureates' affiliations over time.

Table OA5: Average institutional ranking of prize-winners' affiliations, by field and ranking system

Academic Field	U.S. News Best Colleges 2024		U.S. News Best Global Universities 2024–2025		THE World University Rankings 2024	
	Mean	Rank	Mean	Rank	Mean	Rank
	Natural Sciences					
Mathematics	17.9	2	35.6	2	62.5	2
Physics	21.5	3	44.5	11	79.1	7
Chemistry	25.9	9	41.4	8	75.5	4
Life Science	22.8	4	41.2	7	77.9	5
Astronomy	24.9	6	40.5	6	62.7	3
Earth Science	32.2	13	37.6	3	83.1	8
Engineering						
Electrical Engineering	23.9	5	47.7	13	91.2	10
Civil Engineering	25.9	8	61.7	18	224.8	18
Energy Engineering	25.1	7	40.3	5	205.2	17
Environmental Engineering	35.1	14	54.2	17	150.3	16
Materials Engineering	30.9	11	41.5	9	84.0	9
Mechanical Engineering	28.4	10	51.0	16	79.1	6
Social Sciences						
Economics	15.1	1	30.0	1	57.7	1
Political Science	40.2	16	38.9	4	117.0	12
Sociology	31.6	12	41.8	10	120.0	14
Law	46.3	18	50.6	15	129.8	15
Education	40.3	17	45.9	12	118.2	13
Psychology	39.3	15	48.9	14	99.4	11

Notes: Each pair of columns reports the mean ranking of the institutions with which prize-winners in a field were affiliated (*Mean*) and that field's position among the 18 fields (*Rank*, 1 = most prestigious affiliations). The sample is affiliations from 1960 onward and up to 40 years after the PhD, with year 0 the PhD-granting institution; means are averaged across years since PhD. U.S. News Best Colleges 2024 (published September 2023) ranks 439 U.S. institutions on undergraduate criteria, whereas U.S. News Best Global Universities 2024–2025 and THE World University Rankings 2024 (published September 2023) rank research universities worldwide. Institutions absent from a ranking are excluded from its mean; because university rankings do not cover research institutions such as Bell Labs, the National Institutes of Health and CERN, this omits 2.2% of affiliation-years in economics but 24.6% in physics and 31.0% in the life sciences.

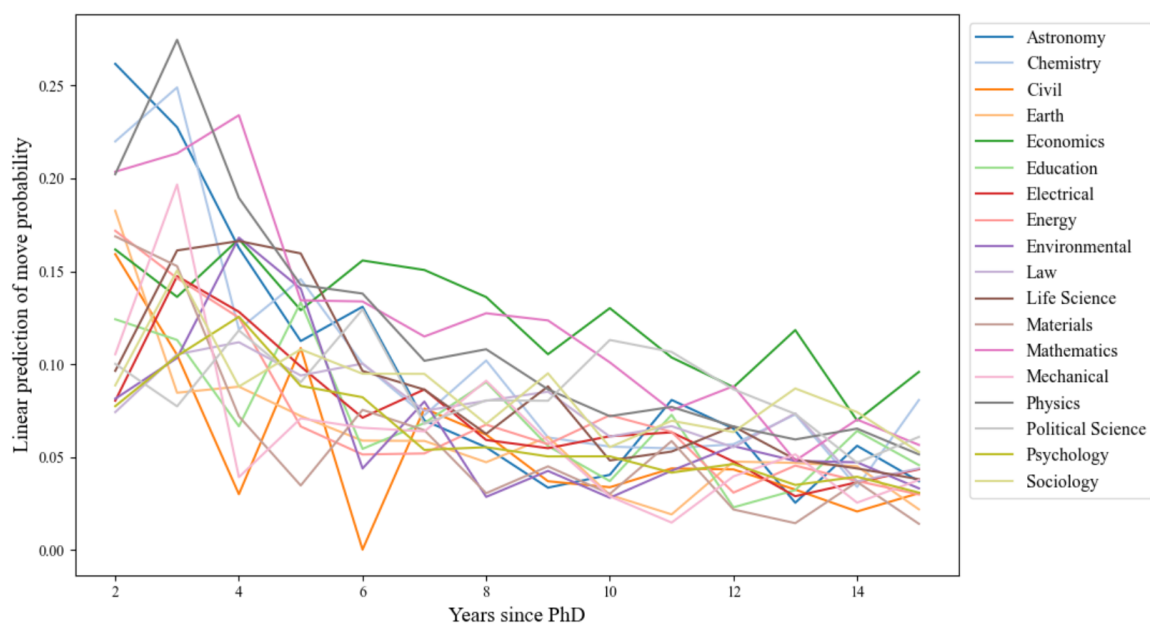
Figure OA7: Normalized HHI of Clarivate Highly Cited Researchers



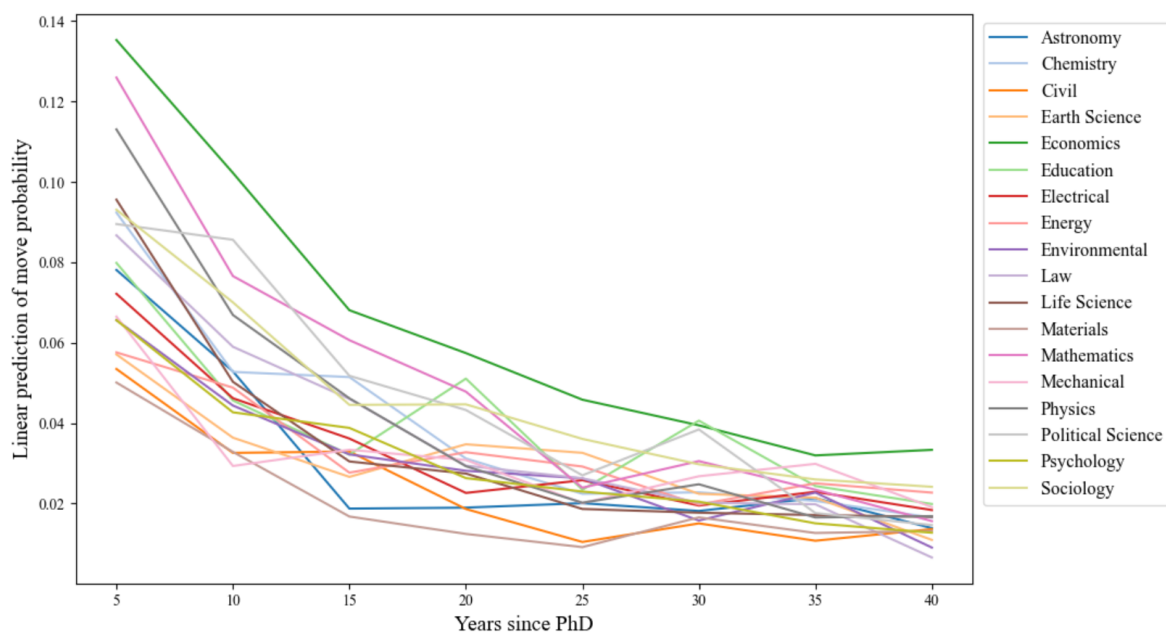
Notes: Normalized HHI values are multiplied by 10,000. The sample covers [Clarivate's public Highly Cited Researchers lists](#) for 2014–2025. Earlier Highly Cited Researchers lists were released irregularly, used different criteria and are not fully preserved in Clarivate's current archive following Thomson Reuters' rebranding as Clarivate. Clarivate identifies researchers with multiple papers ranking in the top 1% of citations within their field and publication year in the Web of Science over the preceding decade. HHI calculations include both primary and secondary affiliations. Clarivate's field categories are renamed, combined and in some cases manually separated to align with the paper's 18-field classification.

Figure OA8: Margins plot of predicted average move probability by subject for prizewinners

(1) Early Career

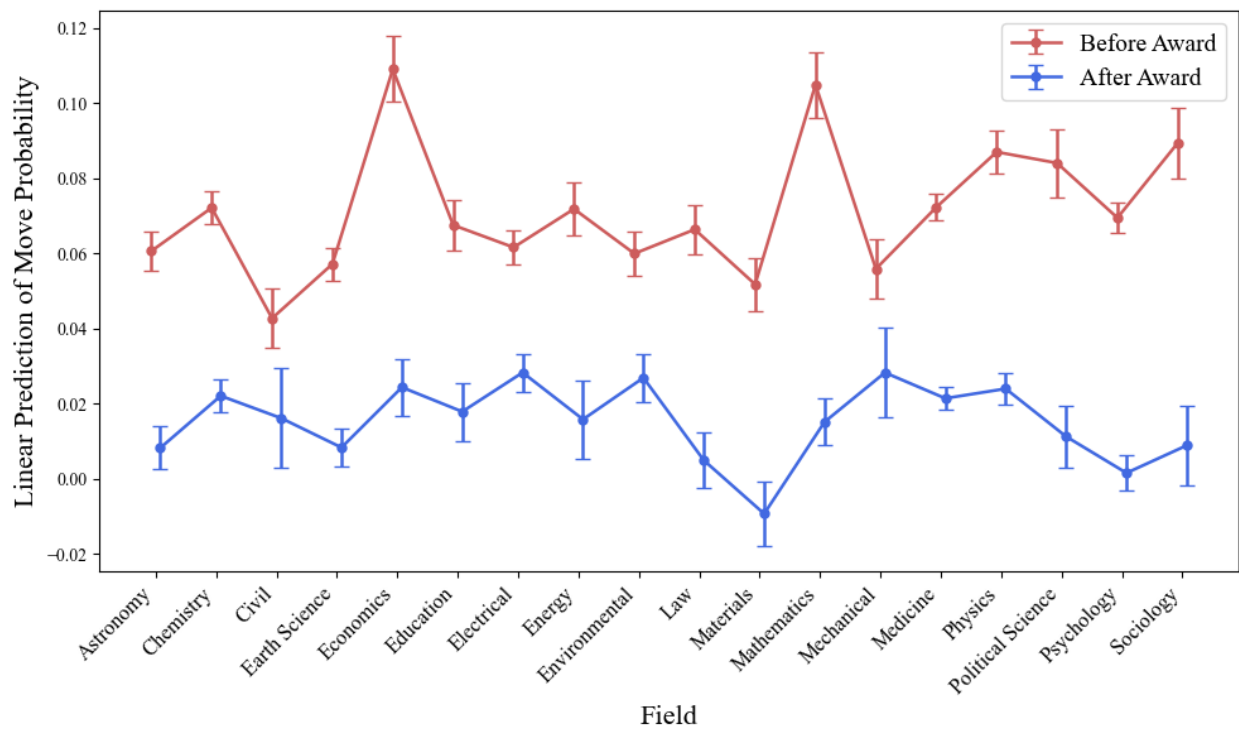


(2) Full Career



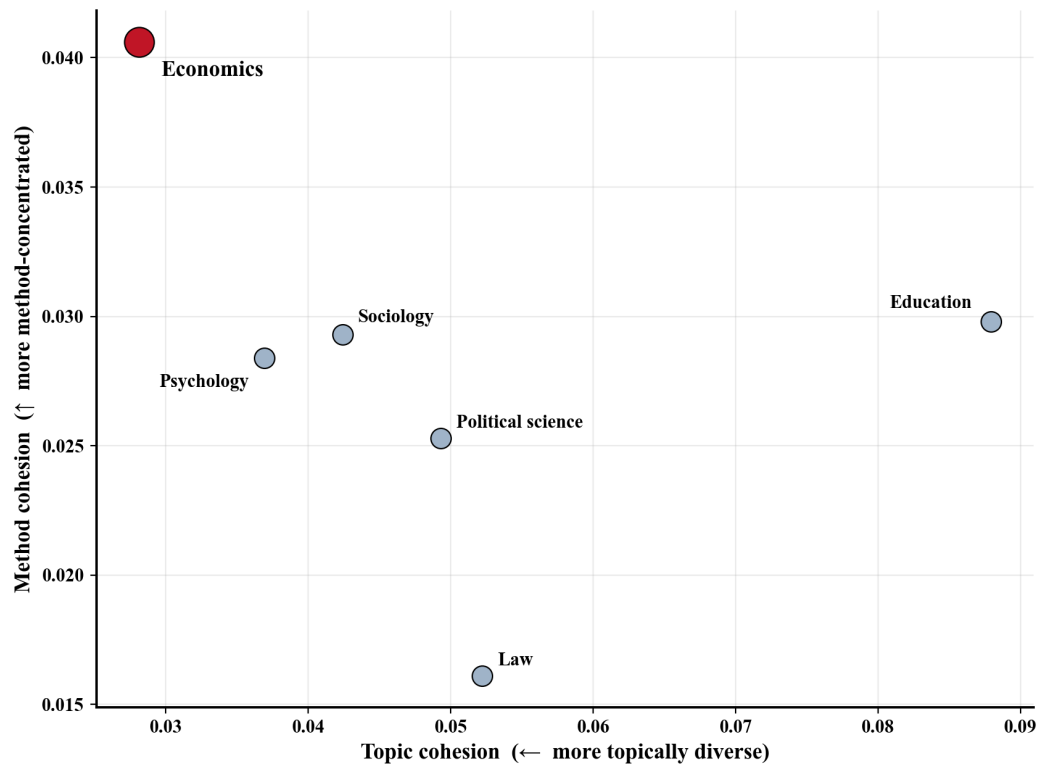
Notes: Early career defined as within 15 years of PhD graduation. Years since PhD graduation larger than 40 years omitted from the graph. Margin's plot from linear probability regression controlling for year fixed effects and years since PhD. Standard errors are clustered at individual level.

Figure OA9: Average annual moving probabilities of award-winning researchers by field:
Before and after the award



Notes: Coefficients measure the average annual probability of institutional mobility before and after the award. Estimates come from a linear probability regression of annual mobility on field and post-award interactions, controlling for year fixed effects and years since PhD. Standard errors are clustered at the individual level.

Figure OA10: Method cohesion versus topic cohesion



Notes: Each point plots one field's topic cohesion (horizontal axis) against its method cohesion (vertical axis); both measures are constructed as described in the notes to Figure 7. Fields toward the upper left combine uniform research methods with diverse subject matter. Economics (red) has the most concentrated methodology with the broadest range of topics.

Table OA6: Hiring move probability of award winners

Panel A: Affiliation at time of the award			
Discipline	Upward (%)	Parallel (%)	Downward (%)
Astronomy	38.22	27.75	34.03
Chemistry	39.27	28.54	32.20
Civil	29.41	38.24	32.35
Earth Science	40.17	29.71	30.13
Economics	33.86	41.04	25.10
Education	32.41	32.87	34.72
Electrical	29.89	26.94	43.17
Energy	22.70	25.53	51.77
Environmental	37.06	21.68	41.26
Law	42.77	23.70	33.53
Materials	36.86	26.81	36.32
Mathematics	26.79	25.32	24.03
Mechanical	22.58	33.87	43.55
Medicine	35.84	23.47	40.68
Physics	37.26	25.00	37.74
Political Science	42.38	24.50	33.11
Psychology	45.17	23.03	31.80
Sociology	50.65	24.03	25.32
Panel B: Affiliation of most time spent			
Discipline	Upward (%)	Parallel (%)	Downward (%)
Astronomy	37.74	29.25	33.02
Chemistry	38.26	29.13	32.61
Civil	25.00	41.67	33.33
Earth Science	41.18	28.17	30.65
Economics	33.99	40.32	25.69
Education	32.40	33.20	34.40
Electrical	26.17	28.35	45.48
Energy	21.59	26.70	51.70
Environmental	36.36	24.24	39.39
Law	40.82	27.55	31.63
Materials	34.64	27.45	37.91
Mathematics	28.51	28.93	42.56
Mechanical	24.75	35.64	39.60
Medicine	34.82	24.16	41.03
Physics	37.26	25.00	37.74
Political Science	42.38	24.50	33.11
Psychology	46.20	21.86	31.94
Sociology	48.15	24.69	27.16

Notes: Records the mobility of prize-winning researchers by comparing the ranks of their employment institutions to those of their PhD granting institutions. Panel A reflects the institution's rank at the time of the award, while Panel B captures the rank of the institution where the majority of the researcher's career is spent. Rankings are based on the number of recipients, and to minimize variability from small sample sizes, ranks were grouped into five-tier clusters.

Table OA7: Field combination of 18 academic disciplines

Discipline	Fields
Natural Sciences	
Mathematics	Statistics, Mathematics
Physics	Physics
Chemistry	Chemistry
Life Science	Pathology, Neuroscience, Molecular biology, Microbiology, Evolutionary biology, Cell biology, Biostatistics, Biophysics, Biomedical engineering, Biological sciences, Biochemistry, Anatomy, Physiology, Pharmacy, Pharmacology, Pharmaceutical sciences, Immunology, Genetics, Epidemiology
Astronomy	Astronomy
Earth Science	Geography, Geology, Marine sciences, Atmospheric science, Meteorology
Engineering	
Electrical Engineering	Computer science, Computer engineering, Electrical engineering, Information science, Information technology, Operations research, Systems engineering
Civil Engineering	Civil engineering
Energy Engineering	Chemical engineering, Natural resources
Environmental Engineering	Ecology, Environmental engineering, Environmental sciences, Environmental health sciences
Materials Engineering	Materials engineering
Mechanical Engineering	Aerospace engineering, Industrial engineering, Mechanical engineering
Social Sciences	
Economics	Economics, Finance, Agricultural economics
Political science	Political science, International affairs
Sociology	Sociology, Gender studies
Law	Criminal justice and criminology
Education	Counselor education, Curriculum & Instruction, Education, Education administration, Special education
Psychology	Psychology

Table OA8: Rank change and assortativity by domain and field

Domain / Field	Upward (%)	Parallel (%)	Downward (%)	Assortativity
Applied Sciences				
Accounting	9	6	85	0.4383
Agronomy	21	29	50	0.0799
Animal Sciences	23	20	57	0.1753
Architecture	7	15	78	0.4887
Business Administration	10	14	76	0.2580
Finance	6	4	90	0.4808
Food Science	14	24	62	0.2361
Horticulture	20	26	53	0.1357
Management	11	8	81	0.3054
Management Information Systems	8	13	79	0.4701
Marketing	9	4	87	0.4450
Plant Science	13	28	59	0.3242
Soil Science	16	27	56	0.1954
Urban & Regional Planning	10	15	75	0.4429
Education				
Counselor Education	14	27	59	0.3161
Curriculum & Instruction	13	20	67	0.3174
Education	11	29	60	0.3475
Education Administration	11	19	70	0.3607
Special Education	9	35	57	0.2607
Engineering				
Aerospace Engineering	10	19	71	0.3199
Agricultural Engineering	17	17	66	0.2273
Civil Engineering	12	12	77	0.3561
Electrical Engineering	12	14	74	0.4082
Environmental Engineering	12	13	74	0.3295
Industrial Engineering	11	13	76	0.3028
Materials Engineering	11	14	75	0.3406
Mechanical Engineering	11	11	78	0.3112
Operations Research	13	14	73	0.4024
Systems Engineering	8	26	66	0.3701
Humanities				
Art History & Criticism	7	5	88	0.4439
Asian Languages	11	9	80	0.3339
Asian Studies	14	8	78	0.2196
Classics & Classical Languages	5	3	92	0.4631
Comparative Literature	9	7	84	0.3985
English Language & Literature	10	4	86	0.3718
French Language & Literature	8	6	86	0.4379
Germanic Languages & Literature	12	4	84	0.3486
History	8	4	89	0.3420
Linguistics	12	10	78	0.4088
Music	8	7	85	0.3552
Near & Middle Eastern Languages & Cultures	17	12	72	0.3518
Philosophy	8	4	88	0.4235
Religious Studies	5	15	80	0.4023
Slavic Languages & Literatures	7	6	88	0.3555
Spanish Language & Literatures	11	7	82	0.2447
Theater Literature, History & Criticism	7	15	78	0.4884
Theological Studies	11	36	53	0.4360
Math & Computing				
Computer Engineering	13	16	71	0.3732
Computer Science	12	8	80	0.4184
Information Science	14	17	69	0.3698
Information Technology	7	23	70	0.3998
Mathematics	9	4	87	0.4142
Statistics	7	7	86	0.4617

Domain / Field	Upward (%)	Parallel (%)	Downward (%)	Assortativity
Medicine & Health				
Communication Disorders & Sciences	10	25	64	0.3651
Environmental Health Sciences	8	42	50	0.3369
Epidemiology	12	32	56	0.1446
Exercise Science, Kinesiology, Rehab, Health	16	26	58	0.1675
Genetics	14	27	59	0.3581
Health, Physical Education, Recreation	9	24	67	0.2770
Human Development and Family Sciences	13	21	66	0.3124
Immunology	18	22	60	0.2030
Nursing	14	34	52	0.2123
Nutrition Sciences	16	25	59	0.3222
Pharmaceutical Sciences	11	36	53	0.3371
Pharmacology	20	21	59	0.2058
Pharmacy	13	45	42	0.2136
Physiology	16	21	63	0.3114
Public Health	15	27	58	0.1613
Social Work	14	14	71	0.3018
Speech & Hearing Sciences	10	25	65	0.4358
Veterinary Medical Sciences	20	36	45	0.1210
Natural Sciences				
Anatomy	10	34	56	0.3037
Astronomy	9	10	82	0.3323
Atmospheric Sciences & Meteorology	12	18	70	0.3683
Biochemistry	15	10	75	0.2521
Biological Sciences	11	8	80	0.2859
Biomedical Engineering	16	22	62	0.2888
Biophysics	11	34	55	0.4104
Biostatistics	11	24	65	0.2857
Cell Biology	17	16	67	0.2961
Chemical Engineering	11	8	81	0.4416
Chemistry	11	6	83	0.3232
Ecology	16	16	67	0.2831
Entomology	21	18	61	0.2025
Environmental Sciences	12	20	68	0.2232
Evolutionary Biology	16	14	71	0.3403
Forestry & Forest Resources	17	33	51	0.2220
Geology	12	9	80	0.3538
Marine Sciences	13	25	62	0.3495
Microbiology	17	17	66	0.2079
Molecular Biology	12	14	74	0.3255
Natural Resources	14	32	54	0.1910
Neuroscience	16	17	68	0.2945
Pathology	23	25	52	0.0999
Physics	10	8	81	0.3360
Plant Pathology	19	19	62	0.2086
Social Sciences				
Agricultural Economics	13	18	69	0.3561
Anthropology	11	5	84	0.3623
Criminal Justice & Criminology	8	11	80	0.3718
Economics	6	91	3	0.4883
Educational Psychology	13	23	64	0.3408
Gender Studies	8	12	79	0.3702
Geography	12	9	79	0.2852
International Affairs	7	17	75	0.3326
Political Science	10	4	86	0.4465
Psychology	15	8	77	0.3467
Sociology	9	5	86	0.4614

Table OA9: Normalized HHI by domain and subfield

Domain / Field	Top 5	Top 10	Top 30	Top 50	Top 100	All
Applied Sciences						
Accounting	385	212	145	123	90	85
Agronomy	263	265	–	–	–	227
Animal Sciences	423	327	222	198	–	195
Architecture	553	428	231	168	–	154
Business Administration	546	358	195	137	–	127
Finance	508	435	273	182	117	94
Food Science	354	236	166	141	–	141
Horticulture	403	305	–	–	–	241
Management	862	586	269	201	134	100
Management Information Systems	395	283	188	101	–	89
Marketing	320	218	134	105	88	82
Plant Sciences	220	220	158	–	–	157
Soil Science	257	156	164	–	–	164
Urban & Regional Planning	576	504	281	176	–	167
Education						
Counselor Education	133	152	97	76	–	73
Curriculum & Instruction	340	217	166	135	101	99
Education	522	403	164	118	–	94
Education Administration	230	207	186	159	95	87
Special Education	342	202	126	–	–	97
Engineering						
Aerospace Engineering	1115	614	256	193	–	170
Agricultural Engineering	644	261	175	139	–	112
Civil Engineering	783	432	294	234	178	151
Electrical Engineering	1317	1011	450	314	189	134
Environmental Engineering	715	512	262	225	166	156
Industrial Engineering	295	305	234	181	–	162
Materials Engineering	719	419	245	186	–	130
Mechanical Engineering	995	603	337	271	195	154
Operations Research	513	413	186	134	–	132
Systems Engineering	357	257	178	134	–	133
Humanities						
Art History & Criticism	503	515	449	385	261	262
Asian Languages	511	484	372	–	–	352
Asian Studies	501	528	402	–	–	400
Classics & Classical Languages	510	428	362	345	202	201
Comparative Literature	475	603	324	–	–	252
English Language & Literature	475	428	293	242	177	111
French Language & Literature	240	239	275	212	–	209
Germanic Languages & Literature	120	154	220	204	–	204
History	526	525	409	363	310	215
Linguistics	555	591	381	225	–	195
Music	229	252	242	208	178	177
Near & Middle Eastern Language & Cultures	374	418	323	–	–	313
Philosophy	661	553	353	281	185	132
Religious Studies	696	567	378	246	202	201
Slavic Languages & Literatures	266	415	284	291	–	299
Spanish Language & Literature	288	277	175	160	–	150
Theatre Literature, History & Criticism	662	458	245	152	131	131
Theological Studies	348	321	194	–	–	147
Math & Computing						
Computer Engineering	345	380	273	200	143	117
Computer Science	1115	871	498	343	206	119
Information Science	247	161	135	105	–	98
Information Technology	179	121	90	68	–	68
Mathematics	501	436	322	256	181	107
Statistics	635	548	306	204	110	99

Domain / Field	Top 5	Top 10	Top 30	Top 50	Top 100	All
Medicine & Health						
Communication Disorders & Sciences	322	232	155	102	–	104
Environmental Health Sciences	210	138	86	82	–	82
Epidemiology	841	524	308	248	–	207
Exercise Science, Kinesiology, Rehab, Health	265	143	191	128	90	80
Genetics	250	165	74	–	–	73
Health, Physical Education, Recreation	343	230	136	111	–	109
Human Development and Family Sciences	208	154	125	110	–	98
Immunology	302	173	80	63	–	49
Nursing	731	380	204	133	87	73
Nutrition Sciences	213	182	127	105	–	88
Pharmaceutical Sciences	310	191	115	72	–	63
Pharmacology	61	78	71	54	–	42
Pharmacy	483	308	171	–	–	116
Physiology	116	112	90	67	47	47
Public Health	371	407	275	197	136	131
Social Work	341	373	224	183	136	127
Speech & Hearing Sciences	263	208	152	115	–	113
Veterinary Medical Sciences	342	340	188	–	–	174
Natural Sciences						
Anatomy	139	69	39	38	–	38
Astronomy	445	329	228	176	–	127
Atmospheric Sciences & Meteorology	497	381	177	138	–	138
Biochemistry	278	224	157	121	89	70
Biological Sciences	165	180	158	130	92	71
Biomedical Engineering	202	202	138	100	–	81
Biophysics	143	139	72	–	–	60
Biostatistics	405	355	225	199	–	176
Cell Biology	219	263	124	100	–	62
Chemical Engineering	467	400	332	282	167	139
Chemistry	607	639	334	267	191	110
Ecology	195	154	128	–	–	116
Entomology	217	167	178	–	–	173
Environmental Sciences	260	219	119	102	87	87
Evolutionary Biology	284	192	142	–	–	140
Forestry & Forest Resources	251	237	163	–	–	163
Geology	685	461	288	221	149	110
Marine Sciences	252	228	130	–	–	122
Microbiology	189	158	99	74	53	51
Molecular Biology	228	242	143	95	–	67
Natural Resources	229	143	119	–	–	113
Neuroscience	277	216	112	81	–	58
Pathology	110	73	37	39	–	39
Physics	503	462	313	266	181	120
Plant Pathology	302	255	206	–	–	204
Social Sciences						
Agricultural Economics	381	289	240	–	–	228
Anthropology	319	302	285	266	192	149
Criminal Justice & Criminology	186	127	105	102	–	110
Economics	947	806	498	375	251	153
Educational Psychology	185	149	117	92	–	88
Gender Studies	223	166	143	138	–	135
Geography	200	213	149	119	101	101
International Affairs	356	360	216	–	–	179
Political Science	861	731	459	370	242	154
Psychology	259	212	183	137	108	70
Sociology	480	421	359	302	192	136

Table OA10: Program prestige and academic placement: economics is the outlier

	First job	10 years out	Ever
<i>Panel A.</i> Economics: the academia–prestige gradient (within field)			
Move up 1 program (pp)	−0.13*** (0.01)	−0.13*** (0.01)	−0.09*** (0.01)
Move up 10 programs (pp)	−1.32*** (0.13)	−1.32*** (0.13)	−0.92*** (0.12)
Best vs. worst program (pp)	−19.1*** (1.9)	−19.1*** (1.9)	−13.3*** (1.7)
<i>N</i> (economics)	11,408	10,980	11,408
<i>Panel B.</i> Comparison across fields (per 10 percentiles up the ladder)			
Other 17 fields (pooled)	+0.12*** (0.04)	−0.00 (0.04)	+0.10** (0.04)
Economics	−1.86*** (0.18)	−1.94*** (0.19)	−1.29*** (0.17)
Economics – other fields (differential)	−1.98*** (0.19)	−1.94*** (0.19)	−1.39*** (0.18)
<i>N</i> (all fields)	267,421	258,352	267,421
Field FE	Yes	Yes	Yes
Graduation-year FE	Yes	Yes	Yes

Notes. Sample: U.S. PhD recipients (Revelio Labs), 1995–2015 graduating cohorts. The dependent variable equals one if the placement is in academia. “Up” denotes a more prestigious program, so a negative entry means top-program graduates are less likely to be in academia. *Panel A* reports a bivariate linear-probability model of academic placement on program rank within economics; the three rows rescale the same slope (per program, per ten programs, and across the full program ladder). *Panel B* pools all 18 fields in a linear-probability model with field and graduation-year fixed effects, interacting program prestige (measured as a within-field percentile) with an economics indicator; the reported rows are the pooled non-economics slope, the economics slope, and their difference, per ten percentiles moved up the ladder. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table OA11: Move-by-Move Employment Mobility by Field and Career Window

Field	5yr Post-PhD Pre-Award			Pre- Award			Post- Award		
	<i>Up</i>	<i>Par</i>	<i>Dn</i>	<i>Up</i>	<i>Par</i>	<i>Dn</i>	<i>Up</i>	<i>Par</i>	<i>Dn</i>
Astronomy	50.7	4.0	45.3	47.8	3.9	48.3	59.1	5.8	35.1
Chemistry	51.5	6.3	42.2	49.7	6.2	44.1	45.9	4.3	49.8
Civil Eng.	39.4	11.6	49.0	48.5	9.1	42.3	37.5	0.0	62.5
Earth Science	49.3	5.5	45.2	48.5	7.5	44.0	33.7	8.0	58.3
Economics	51.9	3.3	44.9	53.1	3.7	43.3	47.6	2.8	49.6
Education	51.2	8.7	40.2	50.6	11.1	38.3	42.7	9.2	48.1
Electrical Eng.	47.1	1.8	51.1	45.5	2.5	52.1	37.8	9.2	53.0
Energy	46.8	17.0	36.2	43.7	15.7	40.6	35.3	19.0	45.6
Environmental	41.7	16.1	42.1	42.1	20.9	37.0	49.1	12.4	38.5
Law	47.0	13.9	39.0	47.7	12.3	40.0	46.8	14.8	38.4
Materials	44.0	18.0	38.0	40.1	13.2	46.8	50.0	0.0	50.0
Mathematics	58.1	3.6	38.3	54.3	3.6	42.0	50.5	4.5	45.0
Mechanical Eng.	39.0	10.6	50.4	39.4	10.1	50.5	64.3	0.0	35.7
Medicine	46.1	5.4	48.6	48.9	5.3	45.8	39.3	10.7	50.0
Physics	53.8	4.8	41.4	53.2	4.5	42.3	41.9	2.9	55.2
Political Sci.	49.6	11.6	38.8	46.1	10.8	43.1	41.0	14.6	44.4
Psychology	48.8	4.8	46.3	48.1	3.8	48.1	42.0	5.8	52.2
Sociology	47.1	12.9	40.1	45.4	12.8	41.8	42.9	17.2	39.9
Overall	49.3	6.9	43.9	48.7	6.9	44.4	43.5	7.7	48.8

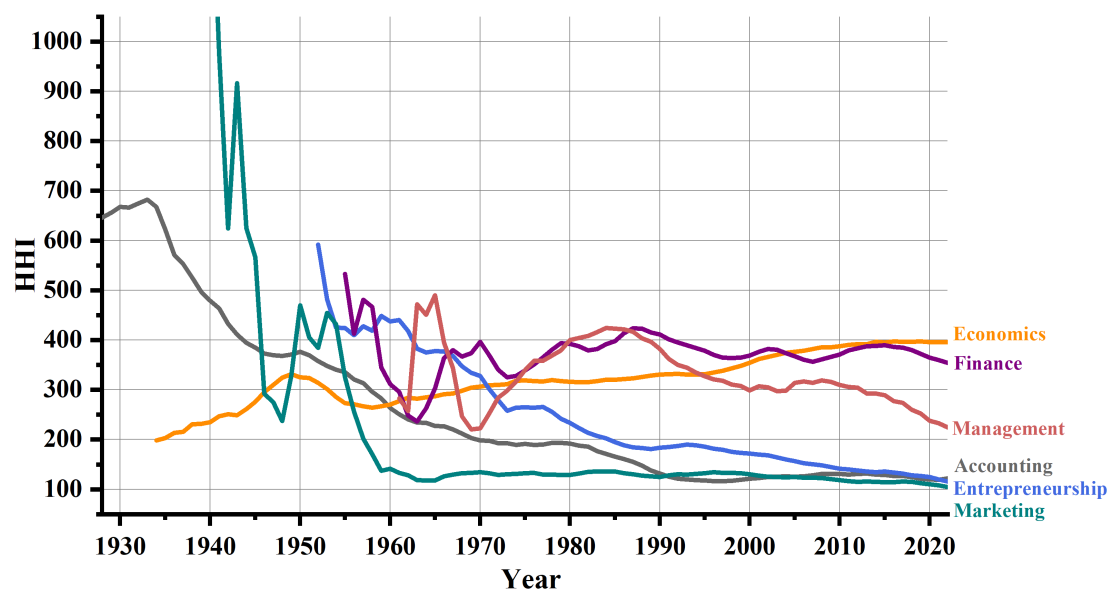
Notes: Each cell reports the field average (across award recipients) of the individual-level percentage of employment moves in that direction within the window. A “move” occurs each time a recipient switches to a different affiliated institution (employment position only). For each recipient, moves within the window are classified as upward (*Up*), parallel (*Par*), or downward (*Dn*) relative to the preceding institution; the recipient’s percentages are computed over their own moves, then averaged across all recipients in the field who make at least one move in that window. Institution ranks are by total unique recipients ever affiliated within each field (dense ranking; rank 1 = most affiliated). “5yr Post-PhD Pre-Award” = moves ≥ 5 years after PhD graduation and before the award year; “Pre-Award” = all moves before the award year; “Post-Award” = all moves from the award year onward. **Shaded row** = Economics.

C Business subfields

Table OA12: List of business awards

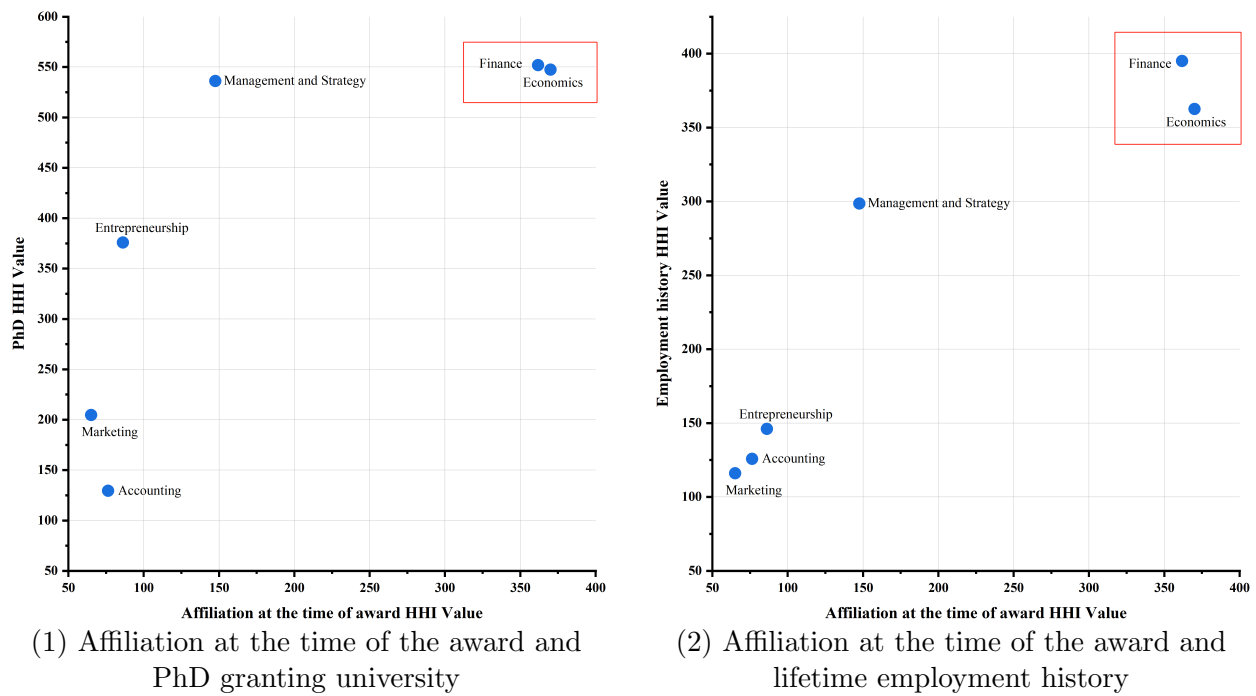
Award name and website	# recipients
Finance	
The Brattle Group Prize in Corporate Finance	46
Deutsche Bank prize in financial economics	6
Fischer Black Prize	10
Onassis Prize	15
Dimensional Fund Advisors Prizes	74
Jensen Prize in Corporate Finance	71
Morgan Stanley - AFA Award for Excellence in Finance	5
Marketing	
American Marketing Association (AMA) Fellows Award	70
Charles Coolidge Parlin Marketing Research Award	88
Paul E. Green Award	84
Weitz-Winer-O'Dell Award	116
Dick Wittink Prize	39
Robert J. Lavidge Global Marketing Research Award	7
AMS Distinguished Fellow Award	56
Accounting	
AAA Distinguished Contributions to Accounting Literature Award	81
Accounting Hall of Fame by the American Accounting Association (AAA)	120
AAA Presidential Scholars	32
AAA Lifetime Service Award	20
Accounting Horizons Best Paper Award	122
Management	
AOM Distinguished Scholarly Contributions to Management Award	9
Herbert Simon Award	18
CK Prahalad Distinguished Scholar-Practitioner Award	14
George R. Terry Book Award	21
Entrepreneurship and Strategy	
Kauffman Prize Medal for Distinguished Research in Entrepreneurship	11
FSF-NUTEK Global Award for Entrepreneurship Research	31
AOM Entrepreneurship Division Foundational Paper Award	29
SMS Fellows	104

Figure OA11: Employment concentration trends for business school subjects



Notes: Normalized HHI reported with ten-year centered moving averages. The subfields are accounting, economics, finance, management, marketing and strategy.

Figure OA12: Normalized HHI for Affiliation at time of award, Lifetime affiliation and PhD affiliation for Business school subjects



Notes: Affiliation at the time of the award is determined based on the institution with which the individual was associated during the year the award was conferred. The PhD-granting institution is identified as the university where the individual obtained their doctoral degree. Lifetime affiliation is computed by considering all institutions with which the individual has been affiliated, with a time-weighted approach to account for the duration of each affiliation.

D Classification of laureates

The classification of Nobel laureates into “broad body of work” or “single testable finding” is determined by both the research content and the language used in the official citation of the award. This dual consideration allows for cases where a laureate’s overall contributions are extensive, yet the award is granted for a distinct, testable finding. Characteristics of each category include:

- **Broad body of work:** Recognized contributions span multiple years, often reflecting a lifetime of work across numerous papers or projects. These awards are typically cited in terms such as “for contributions to or analysis of” a given field or subfield, highlighting a comprehensive influence.
- **Single testable finding:** Awards in this category are based on results from a single study or a limited set of studies produced within a relatively short period. Such citations often specify “for methodological contributions, invention, discovery, experiment, or proof of” a precise finding, underscoring the significance of a discrete result.

To ensure objectivity in our classification, we use large language models (LLMs) to complement human interpretation. Our approach relies on human analysis of past scholarly work and research contributions, with the aid of the LLMs’ classification. We ask ChatGPT-4 the following question:

“Consider the winners of [Prize] in [Year] individually. Would you classify each laureate as having been awarded for a broad body of work or a single testable, specific finding? A broad body of work refers to substantial, generalized advancements in an area or topic. A single testable finding, by contrast, involves a demonstrable theory or empirical discovery, for example, the proof of a theorem, invention of a specific method or object, or development of a disease treatment. Consider the award-winner’s research portfolio, the work for which the prize was awarded, and the wording of the award citation. ”

Below are three examples of classifications made by the LLM, using the Nobel Prize in Chemistry for the years 2019, 2016, and 2013.

2019

The 2019 Nobel Prize in Chemistry was awarded to John B. Goodenough, M. Stanley Whittingham, and Akira Yoshino for the development of lithium-ion batteries. Here’s a breakdown

of their prizewinning work and an assessment of whether it represents a broad body of work or a specific testable finding:

1. John B. Goodenough

Contribution: Goodenough significantly improved the battery’s capacity by identifying cobalt oxide as a material that could hold a higher energy density, making the lithium-ion battery much more powerful and practical.

Category: Broad body of work. While Goodenough’s specific contribution to battery technology is a landmark discovery, his work spans decades and has led to multiple advancements in energy storage. His contributions have been applied broadly across industries and scientific fields.

2. M. Stanley Whittingham

Contribution: Whittingham laid the foundation for lithium-ion batteries in the 1970s by demonstrating how lithium could be used to store energy in batteries. He developed the first functional lithium battery by using titanium disulfide as a cathode, which allowed for lithium ions to be stored between layers of the material.

Category: Testable, specific finding. Whittingham’s work was focused on a specific discovery—the creation of the first rechargeable lithium battery—setting the foundation for further advances in energy storage.

3. Akira Yoshino

Contribution: Yoshino built on Goodenough and Whittingham’s work to develop the first commercially viable lithium-ion battery by using a safer material, petroleum coke, which allowed lithium ions to move safely within the battery. This innovation made the battery ready for widespread use in portable electronics.

Category: Testable, specific finding. Yoshino’s work was a focused improvement of existing lithium-ion battery technology, making it safer and more practical for commercial applications.

Conclusion: The 2019 Nobel Prize in Chemistry recognizes a mix of broad body of work (for Goodenough) and testable, specific findings (for Whittingham and Yoshino). While Whittingham and Yoshino’s contributions represent more targeted breakthroughs in the development of rechargeable lithium-ion batteries, Goodenough’s decades of work on battery technology encompass a broader impact across multiple scientific and industrial applications. Together, their contributions led to one of the most important technological advancements

in modern energy storage.

2016

The 2016 Nobel Prize in Chemistry was awarded to Jean-Pierre Sauvage, Sir Fraser Stoddart, and Bernard L. Feringa for the design and synthesis of molecular machines, which are molecules that can perform mechanical tasks when supplied with energy. Here's an overview of their prizewinning work and whether their contributions represent a broad body of work or a specific testable finding:

1. Jean-Pierre Sauvage

Contribution: In 1983, Sauvage made a crucial breakthrough by linking two ring-shaped molecules together to form a catenane (interlocked molecular rings). This was a key step toward building molecular machines, as it showed that molecules could be linked together mechanically rather than chemically.

Category: Testable, specific finding. Sauvage's discovery of mechanically interlocking molecules represents a clear, testable innovation that laid the foundation for molecular machines. His work was focused on a specific achievement that opened up new possibilities in molecular chemistry.

2. Sir Fraser Stoddart

Contribution: Stoddart developed a rotaxane in 1991, a molecular structure in which a ring is threaded onto a molecular axle and can move along it. He later used this structure to build molecular machines, such as molecular elevators and muscles, demonstrating that molecular structures could perform mechanical tasks.

Category: Testable, specific finding. Stoddart's invention of rotaxanes and subsequent construction of molecular machines are specific, replicable innovations in molecular engineering. His work directly led to the development of functional molecular devices.

3. Bernard L. Feringa

Contribution: Feringa developed the first molecular motor in 1999, a molecular structure that could rotate in one direction when exposed to light. He later refined this design, creating more advanced molecular machines, including molecular nanocars. **Category:** Testable, specific finding. Feringa's creation of the molecular motor is a clear, focused discovery that demonstrated the ability to control mechanical motion at the molecular level, leading to specific applications in nanotechnology.

Conclusion: The 2016 Nobel Prize in Chemistry was awarded for testable, specific findings. Each of the laureates made concrete contributions to the development of molecular machines, achieving specific, measurable advancements in chemistry. Their work demonstrated the possibility of controlling molecular structures and movements, leading to the creation of functional nanotechnology that operates on a molecular scale.

2013

The 2013 Nobel Prize in Chemistry was awarded to Martin Karplus, Michael Levitt, and Arieh Warshel for the development of multiscale models for complex chemical systems, which made it possible to simulate chemical reactions using both quantum mechanics and classical physics. Here's an overview of their prizewinning work and whether it represents a broad body of work or a specific testable finding:

1. Martin Karplus

Contribution: Karplus was a pioneer in developing computational models that combine quantum mechanics and classical physics to simulate the behavior of molecules during chemical reactions. His work laid the groundwork for computer simulations that predict how molecules behave in real-world scenarios.

Category: Broad body of work. Karplus's contributions span decades and involve multiple advancements in computational chemistry. While his work led to specific findings, his broader impact lies in developing methods that are used across chemistry, biology, and materials science for simulating molecular systems.

2. Michael Levitt

Contribution: Levitt made significant contributions to the development of methods for molecular dynamics simulations, which allow scientists to study the motion of atoms and molecules in biological systems over time. His work helped create practical tools for simulating biological processes, such as protein folding.

Category: Broad body of work. Levitt's work involved multiple theoretical and practical advancements in computational biology, making it a broad contribution to the field. His methods have wide applicability in various areas of molecular biology and chemistry.

3. Arieh Warshel

Contribution: Warshel developed quantum mechanics/molecular mechanics (QM/MM) models, which allowed the precise simulation of chemical reactions by treating the active parts of molecules quantum mechanically and the surrounding areas using classical physics.

This method made simulations computationally feasible while retaining accuracy.

Category: Broad body of work. Warshel's development of QM/MM methods contributed broadly to computational chemistry, allowing for the simulation of complex chemical systems. His work has been widely used across multiple fields, from enzymatic reactions to materials science.

Conclusion: The 2013 Nobel Prize in Chemistry recognizes broad bodies of work. Each laureate contributed to the development of computational models that span various applications in chemistry and biology. While their methods led to specific simulations and discoveries, their overall contributions lie in the creation of generalizable tools and models for understanding complex chemical systems at multiple scales, blending quantum mechanics and classical physics. These tools have had a lasting impact on theoretical and computational chemistry, enabling researchers to simulate molecular processes that would otherwise be impossible to observe experimentally.